

Near-extremal Kerr Black Holes and EMRI Waveform Modelling for LISA

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Abstract

Extreme mass ratio inspirals (EMRIs) are gravitational wave (GW) events in which a stellar-mass compact body enters a highly relativistic orbit about a massive black hole (MBH) and slowly inspirals due to the emission of GWs. These events will be prime observational targets for future-generation GW observatories such as the Laser Interferometer Space Antenna (LISA). Accurate and efficient gravitational waveform models are required for detection by LISA and for data analysis purposes. Prior work has shown that quasi-circular EMRIs about an MBH with a near-extremal spin parameter of $a \sim 1$ can constrain the spin parameter of the MBH with extremely high precision. FastEMRI-Waveforms (FEW) is a state-of-the-art toolkit used to rapidly generate adiabatic EMRI waveforms. In this project we leverage the pre-existing architecture of FEW to build an adiabatic waveform model for EMRIs about a near-extremal MBH with spins as high as $a = 1 - 10^{-12}$. Our waveform model also allows for orbital eccentricity as high as $e = 0.4$. This waveform model was used to examine the near-horizon regime of the primary MBH and extend the prior results of constraining the spin parameter to include the effects of orbital eccentricity. We show that the precision to which LISA will be capable of constraining the spin parameter of the primary MBH exhibits a weak dependence on the initial orbital eccentricity of the EMRI. With initial orbital eccentricities of $e_0 = 0.4$ and primary a spin parameter of $a = 1 - 10^{-8}$, we measure $\Delta a \sim 10^{-10}$, which is consistent with the same measurement in the quasi-circular case.

Declaration

I declare that this dissertation was composed entirely by myself.

Chapter 1 provides an introduction to the topic of this dissertation.

Chapter 2 is a literature review and does not contain original research.

Chapter 3 describes how I implemented and tested the model. All data produced for this chapter, and coming chapters was provided by Dr. Zachary Nasipak.

Chapter 4 describes work that was carried out entirely by me. The results of this chapter are a combination of validation of results previously obtained by my primary supervisor Dr. Alexander (Ollie) Burke, and also an original extension of said results which is not available in the current literature.

Calculations carried out during this project were done using Python. The Python package FastEMRIWaveforms was the primary focus of the coding content, and was modified to produce the waveform model of this project. My co-supervisor Dr Zachary Nasipak produced all gravitational wave flux and amplitude data using his PYBHPT Python package.

Personal Statement

I spent the first 2 weeks of the project familiarising myself with the FastEMRIWaveforms codebase, as this was where I would be spending a lot of my time. Alongside this familiarisation process, I had meetings with my supervisors to discuss what the data grid of gravitational wave flux and amplitude data should look like, and what regions of parameter space would be covered.

After these two weeks, I implemented the first generation of the interpolation grid into FEW. Provided with gravitational wave flux data by Dr. Nasipak, I incorporated this data into the codebase to complete the near-extremal trajectory module. This took two weeks. I then spent an additional week testing the behaviour of the trajectories, found and fixed bugs in my code, and compared the interpolated flux data to other waveform models. It was decided that a denser data grid was needed to better model the behaviour of the trajectories near the horizon of the primary black hole. This took up to week five of the project.

With a functioning trajectory generator, I could begin work on the amplitude module. This did not take much work, as the same interpolation grid that was used for the fluxes was also used for the amplitudes. A break from coding was taken as data was generated, and in this time I began work on my dissertation report. Amplitude data became available in week seven.

With the new amplitude data, I began to generate waveforms, and quickly noticed a strange up-tick behaviour in the waveform amplitudes towards the end of the inspiral. I discussed the potential origins with my supervisors. We originally tried to fix the behaviour by computing more amplitude modes, but the up-tick remained. Upon further investigation, and comparison with the FastEMRIWaveforms' eccentric waveform model, the behaviour was found to be a result of interpolation error. The amplitudes were then computed on a denser grid to remove the unphysical behaviour and increase accuracy.

I was now in the final week of July. The next step was to compute Fisher matrices with the waveform model, and to study the detectability of the signals. I built a code to take derivatives of the waveform, and compute the inverse Fisher matrices. The results were obtained by the end of July, and the rest of my time was spent writing this dissertation.

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Chapter 1

Introduction

One of the most studied and astrophysically relevant consequences of general relativity (GR) is the existence of black holes (BHs). Due to the light-swallowing nature of BHs, they are difficult objects to observe with electromagnetic observations. To do so, elaborate observational experiments have been conducted which measure the *effect* that a BH has on nearby matter. Observations of BHs using telescopes can be made, such as the observation of Sagittarius A* at the centre of the Milky Way [1]. This observation was made indirectly via the astrometry of stars orbiting the massive black hole (MBH). Another example is the observation of the BH shadow of the MBH M87* in 2019 [2], observed by the Event Horizon Telescope. This observation allowed for a measurement of the spin parameter $a = J/M$ of the MBH to be made, which was calculated to be $a = 0.9 \pm 0.05 M$ [3].

One key prediction of GR is that of gravitational waves (GWs), derived by Einstein as a result of his gravitational field equations [4]. GWs are produced when any mass is accelerated, but the effect is negligible unless the mass involved is extremely massive and is accelerated to relativistic speeds. With a suitable GW observatory, GWs allow for the observation of compact body binaries (COBs), an astrophysical system in which two compact bodies, such as neutron stars or BHs, orbit each other, emitting GWs as they do so [5].

GWs were first directly detected by the Advanced LIGO GW detector in 2015, where ground-based interferometers measured tiny perturbations in spacetime; gravitational waves emitted by the inspiral and merger of two black holes, with masses of $35^{+5}_{-3}M_{\odot}$ and $30^{+3}_{-4}M_{\odot}$ [6, 7]. Since this initial observation, more GW observatories have been made operational such as VIRGO [8] and KAGRA [9], forming the LIGO-VIRGO-KAGRA (LVK) collaboration. Data from these detectors have been released in five catalogues of GW events [10, 11, 12, 13, 14], allowing for the testing of fundamental physics [15].

While ground-based detectors such as LIGO have revolutionised GW astronomy, the frequency range of the terrestrial interferometers can only detect GWs with a frequency somewhere in the range of $10 - 10^4$ Hz, bounded below by seismic activity and weather

systems on Earth [16]. Therefore, to observe in a lower frequency range, and thus observe more massive GW events, a larger and more stable detector is required, free from Earthly constraints. To solve this problem, the Laser Interferometer Space Antenna (LISA) has been officially adopted by the European Space Agency (ESA) as a large-class mission, and is due to launch in the mid-2030s [17].

A key GW source that LISA will observe is the *extreme mass ratio inspiral* (EMRI), which occurs when a stellar-mass compact body with mass μ enters a highly relativistic orbit about an MBH of mass M , with a resultant mass ratio of $\mu/M = \epsilon \leq 10^{-4}$. The smaller black hole, henceforth referred to as the *secondary*, has a generic orbit about the MBH, henceforth referred to as the *primary*, and slowly loses orbital energy and angular momentum in the form of GWs. The frequencies of these GWs are in the millihertz band, which LISA will be able to detect [18].

The GW signal produced by EMRIs will be in LISA’s sensitive frequency band for up to several years. To extract the GW signal from the background noise in the LISA datstream, a large catalogue of gravitational waveform templates is required for a matched filtering search [19]. These waveforms have to be extremely efficient to compute for use in any form of matched filtering or global fit algorithm. The waveforms from EMRIs that LISA will detect are very sensitive to small changes in the system parameters and as a result they can be used to make high-precision measurements of the system parameters [20]. Another consequence of this sensitivity is that the waveform models used in the matched filtering algorithms must be highly accurate as well as efficient. Recent developments in the field of gravitational self-force theory (GSF) and EMRI waveform modelling allow for codes that can rapidly generate EMRI waveforms for detection with LISA and data analysis [21, 22]. One such codebase is FastEMRIWaveforms (FEW) [23, 24], which will be integral to the work done in this project.

The conjectured upper limit for BH spin is $|a|=1$, as any spin higher than this would result in a naked singularity, which would break the cosmic censorship conjecture, which states that singularities must always be covered by an event horizon [25]. The *Thorne limit* [26] attempts to set an astrophysical upper limit of $|a| \lesssim 0.998$, which comes from proving that this is the maximum spin that can be reached via standard accretion processes. However, other viable astrophysical methods of increasing the angular momentum past this limit have been studied [27, 28]. While no evidence for near-extremal BHs has been found in LVK data thus far, recent work in Ref. [29] shows that should near-extremal MBHs exist, if LISA were to detect a GW signal from an EMRI about one, the waveform and frequency signature from the signal would constitute a *smoking gun*, meaning that the GW *must* have come from an EMRI about a near-extremal MBH. Work in Ref. [30] shows that an EMRI around a near-extremal BH allows for extremely precise measurements of the MBH’s spin parameter, with roughly 3 – 4 orders of magnitude higher than the non-extremal case. However, this analysis was conducted for quasi-circular inspirals, and thus neglected the effects of orbital eccentricity and inclination in the trajectory of an EMRI.

The primary objective of this project is to utilise the existing FEW architecture to build a waveform model for near-extremal EMRIs, which includes orbital eccentricities up to $e = 0.4$, and primary BH spin of up to $a = 1 - 10^{-12}$. This waveform model can then be used to explore the characteristic near-horizon behaviour of EMRI waveforms discussed in Ref. [29]. The model can then also be used for a Fisher matrix analysis, where we measure the amount of information that the waveform holds about each parameter of the EMRI system, and thus estimate the precision to which LISA can measure each parameter. This extends the results of Ref. [30] to include eccentricity.

Chapter 2

Background Theory & Formalism

2.1 Gravitational Wave Astronomy

This subsection examines the limitations of the current generation of ground-based GW observatories, and the implications these limitations have on the observable GW sources. The capabilities of future generation space-based GW observatories —such as LISA— will be examined, again with the motivation of introducing the sources of GWs that future observatories aim to detect, one of which is known as an *extreme mass ratio inspiral*, which will be discussed in detail in section 2.1.2. The modelling of these EMRIs will be the motivation of the remainder of this dissertation.

2.1.1 Gravitational Wave Observatories

GW detectors, whether they are space-based or ground-based, operate on the same underlying principle; laser interferometry, which was broadly developed for the purpose of GW detection by Rainer Weiss in the 1970s. This technique allows for tiny changes in the length of a laser beam to be measured. GWs manifest in spacetime far from the GW source as infinitesimal perturbations to spacetime, or as a GW observatory sees it, as small periodic changes in the length of the laser beam. These tiny changes in length are described as a metric perturbation to flat spacetime. In an appropriate gauge, the metric perturbation $h_{\mu\nu}$ can be shown to obey a wave equation, and thus have plane-wave solutions of the form,

$$\bar{h}_{\mu\nu}(\vec{x}, t) = \text{Re} \left(\int d^3k A_{\mu\nu}(\vec{k}) e^{i(\vec{x} \cdot \vec{k} - \omega t)} \right), \quad (2.1)$$

as derived in Appendix A. In GR, GWs have two polarisations which perturb spacetime in a purely spatial way, i.e. only affect proper distance,

$$h_{xx}^{TT} = -h_{yy}^{TT} \equiv h_+(t - z) \quad (2.2)$$

$$h_{xy}^{TT} = h_{yx}^{TT} \equiv h_\times(t - z). \quad (2.3)$$

Here the superscript TT indicates that the transverse traceless gauge (also described in section A) has been adopted. The four terms shown above are the only non-zero terms of

the tensor $h_{\mu\nu}$. The effect that this perturbation has on the proper distance L between two freely falling particles, separated along the x -axis, is given by [31],

$$\frac{\delta L}{L} \simeq \frac{1}{2} h_{xx}^{TT}(t, z = 0). \quad (2.4)$$

This is the periodic change in length that a GW observatory measures. The polarisations h_+ and $h_\times$ will henceforth be referred to as the GW *strain*. This strain is what GW observatories are built to detect, and it is what gravitational waveform models, such as the one built for this dissertation, aim to model as accurately as possible.

Current & Future Generations of GW Observatories

The first GW observatories to directly detect a GW were the aLIGO observatories in Washington and Louisiana, both of which detected GW event GW150914 [6]. As noted in the introduction, the GW event detected was the coalescence of two black holes with masses of $35_{-3}^{+5}M_\odot$ and $30_{-4}^{+3}M_\odot$, which coalesced to form a single remnant BH with a mass of $61.5_{-2.9}^{+3.4}M_\odot$, meaning approximately $3M_\odot$ was radiated away in the form of gravitational radiation. The dimensionless spin parameters of the two BHs before merging were constrained to be < 0.65 and < 0.75 respectively, at the 90% credible interval [7].

Since this initial observation, a total of 391 GW events have been detected and are likely to be of an astrophysical origin [14]. All of these events consist of compact object (CO) mergers, where the masses of the COs in any given merger are typically on the same order of magnitude of $\sim (1 - 200)M_\odot$. These events involve a rapid inspiral of two COs driven by the loss of energy and angular momentum carried away from the system in the form of GWs, some of which are detected by terrestrial detectors. The GW signal from the inspiral is typically only in LIGO's operational frequency band for the final minutes of the inspiral. By this late stage the binary has circularised, a process by which eccentricity is radiated away via GWs [32]. The shorter observational time of these binaries results in large uncertainties on the system's mass and angular momentum of the involved COs. Aside from larger-than-ideal uncertainties, ground-based observatories also have a hard lower bound on the frequency range that they can observe in, which, as will soon be shown, restricts the total mass of the GW source to be $\lesssim 100M_\odot$ and thus motivates a space-based GW observatory. This hard lower bound is a consequence of noise caused by seismic activity, which is strongest at low frequencies where it cannot be avoided [16]. This seismic lower bound can be avoided entirely by space-based observatories such as LISA.

LISA is currently due to launch in the mid-2030s, and is expected to have a nominal mission duration of $4\frac{1}{2}$ years [17]. During this time LISA will observe hundreds of thousands, or even millions of GW signals, a lot of them simultaneously. Tens of thousands of these events are expected to be resolvable, with the parameters of each system potentially being found, while the remaining unresolvable signals will form an astrophysical background, which must be accounted for in a global fit [17, 33]. An overview of the main outline of the mission is shown in figure 2.1. Each aspect of the mission poses a mammoth task,

from the science and engineering required to detect such tiny perturbations of spacetime in space, to extracting individual GW signals from a superposition of waveforms and noise in the LISA data stream. A lot of these problems are not unique to space-based observatories, but are faced by terrestrial observatories too. For both, waveform modelling plays a crucial role. The figures quoted of the masses and spin of the BHs involved in the first ever direct detection of GWs above, were not the values originally reported in the initial report of the detection [34], but instead come from an improved analysis of the data utilising a more accurate waveform model, thus allowing for higher precisions of the measurement of the system parameters [7].

The role of accurate and efficient waveform models is even more crucial for space-based detectors, as they require a global fit, which will simultaneously fit all GW signals in the LISA data stream using accurate waveform models [33]. This global fit requires a catalogue of all potential sources of GWs detectable by LISA, which can then be used to match the individual signals observed in the LISA data stream to the precomputed waveform that best matches it via the process of matched filtering [17, 35, 36]. In this way the parameters of the GW source can be estimated. This task requires the waveform models to be highly efficient. FastEMRIWaveforms is a PYTHON package with this exact purpose in mind [23, 24].

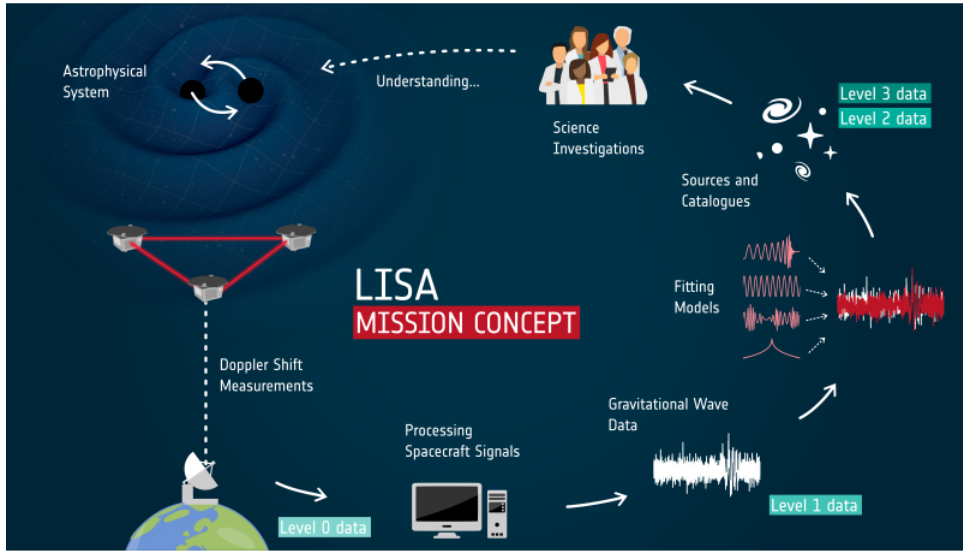


Figure 2.1: From Ref. [17], a high-level overview as to how the LISA mission will be structured. The work in this dissertation focuses on the ‘Fitting Models’ part of the mission concept.

Observational Potential of LISA

It will now be shown with a simple approximate argument, what types of GW events are detectable with LISA. When predicting the astrophysical events that any GW observatory might be expected to observe, it is useful to make rough order of magnitude estimations. Consider an idealised scenario of two BHs, one of mass μ , and the other of mass M ,

orbiting each other in a circular orbit. To get an approximation of their orbital frequency, and thus of the frequency of GWs emitted by the system, we can utilise the Keplerian formula for orbital frequency (with $G=c=1$):

$$\Omega = \sqrt{\frac{\mu + M}{R^3}}, \quad (2.5)$$

where R is the radial distance between the two black holes. We can approximate and assume that R in equation (2.5) is close to the Schwarzschild ISCO of $R \simeq 6(M + \mu) = 6M_{\text{tot}}$. Using $f_{\text{detector}} \simeq \Omega/\pi$ yields,

$$f_{\text{detector}} \simeq \frac{1}{6\pi M_{\text{tot}}} \implies M_{\text{tot}} \simeq \frac{c^3}{6\pi G f_{\text{detector}}} \quad (2.6)$$

$$\simeq \frac{4410}{f_{\text{detector}}} M_{\odot} \quad (2.7)$$

where the second equation has been rearranged for M_{tot} and recovered SI units. This equation gives an approximate mass range for the masses of binaries in GW events that a detector that observes with a frequency of f_{detector} can detect. The frequency range that LISA is sensitive in, as shown in figure 2.2, is roughly $10^{-1} - 10^{-4}$ Hz. Thus, this frequency range in Eq. (2.7), yields a range in M_{tot} of $\sim 10^4 - 10^7 M_{\odot}$.

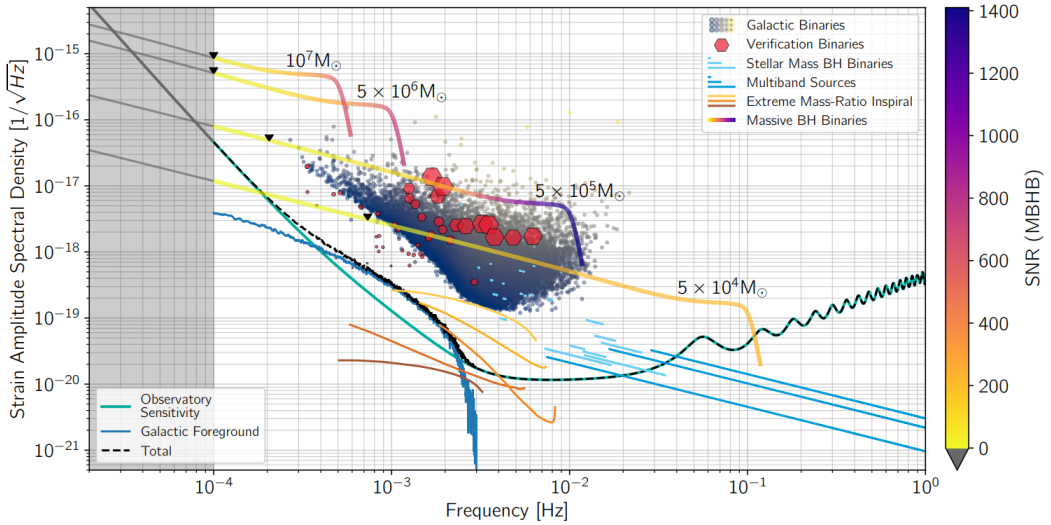


Figure 2.2: From [17], displaying GW sources that lie in the LISA “sensitivity bucket”, which is the dotted black line on the plot (a combination of instrumentation noise, and the unresolved signals from galactic foreground). Any source with a strain amplitude spectral density (ASD) higher than this dotted black line, will be detectable by LISA. The colour bar indicates the SNR for the massive BH binaries plotted. EMRIs lie at the bottom of the LISA bucket, indicated by the thin yellow and orange coloured lines, representing the individual harmonics of an EMRI.

Within this mass range, there are several new astrophysical events that LISA will be capable of observing that are undetectable by terrestrial observatories including, but not limited to:

- ★ Massive black hole binaries (MBHB) — Systems comprised of two MBHs with masses between $10^4 M_\odot$ and $10^7 M_\odot$ such as the MBH at the centre of the Milky Way [37]. As can be seen in figure 2.2, these events can have extremely high SNRs, and can be detected from very high redshifts. Observations of MBHBs will constrain cosmological parameters [38, 39], and help gain an understanding of how MBHs form [40].
- ★ Comparable mass stellar-mass CO mergers — Similar to MBH binaries, but scaled down to masses on the order of $\sim 10 M_\odot$, the same systems terrestrial detectors observe, but LISA will be able to observe the early parts of the inspiral, and thus may observe orbital eccentricity [17].
- ★ Extreme mass ratio inspirals (EMRIs) — The stellar capture of a stellar-mass CO by an MBH, to form a highly mass-asymmetric binary system — The source of interest for this work.

The gravitational observation of MBHs will give unparalleled insight into the geometric structure of their spacetimes, and also their astrophysical environments. The extreme mass ratio inspiral will be especially pivotal, for reasons to be discussed, in the understanding of the physics *and* mathematics of black holes.

2.1.2 Extreme Mass Ratio Inspirals

As noted, an EMRI occurs when a roughly stellar-mass compact object (the *secondary*) enters a highly relativistic orbit about an MBH (the *primary*), with the mass ratio — ϵ — between the primary and secondary being ‘extreme’, or more specifically, $\epsilon \leq 10^{-4}$. Due to this extreme mass ratio, the inspiral evolves *adiabatically*, and thus the inspiral takes an extraordinary amount of time, with an inspiral timescale T_{inspiral} that scales as $\sim M/\epsilon$ [22]. During this inspiral, the secondary is expected to complete $\sim 10^4 - 10^5$ orbital cycles [20]. Compared to the timescale of seconds that LIGO observes for, GWs produced by an EMRI are expected to be in-band for upwards of a year and potentially for the entire lifespan of the mission, before reaching the innermost stable orbit and plunging into the central MBH. The trajectory of the secondary is also expected to be periodic in three fundamental frequencies; azimuthal frequency Ω_ϕ , polar frequency Ω_θ , and radial frequency Ω_r . This is a result of having generic orbits with relatively high eccentricities and inclination, leading to complicated waveforms.

The long inspiral time results in very precise measurements of the parameters of the system, accumulating sufficient signal-to-noise ratios (SNRs) to be observed by LISA, with some potentially reaching higher than ~ 50 [17], well above the estimated LISA threshold of $\sim 20 - 30$ [20]. These GW signals are highly sensitive to changes in the parameters of the EMRI system, and thus allow for highly accurate tests of GR, and allow for precise probing of the primary MBH’s spacetime, and astrophysical environment. The dimensionless Kerr spin parameter, as well as the detector-frame mass of the primary BH can be measured with a predicted error of only $\sim 0.01\%$ [17]. This would allow for testing of the no-hair theorem, which states that the spacetime of a quiescent black hole is fully

described by the Kerr metric [41] (which will be introduced in section 2.2) [42]. In general, deviations from classical GR would manifest themselves in the energy and angular momentum fluxes of the system, which affect the inspiral dynamics, and thus the GW that LISA would detect [43, 44]. Very small differences in fluxes can have a measurable impact on the gravitational waveform, where the phase of the waveform is highly sensitive to small changes in inspiral dynamics.

Aside from the constraints that LISA could place on mathematical theories of gravity, there are naturally also implications for physics. The environments around MBHs that reside at the centre of a galaxy are difficult to probe observationally, with distant galactic nuclei only being visible when they are ‘active’. A smaller stellar-mass black hole would provide a detailed map of any gravitationally-interacting material in the primary MBH’s vicinity, such as an accretion disk, or even dark matter [45]. The expected science that can be done with EMRIs is best summarised in Refs. [39, 43].

Our discussion of EMRIs will be expanded in section 2.4.2, where modelling techniques are discussed, and in section 2.5, where the special case of an EMRI about a near-extremal primary BH is discussed in detail, as this special case is the focus of this project.

2.2 Rotating Black Holes

Any black hole formed via gravitational collapse will have some angular momentum, or ‘spin’, as any small amount of angular momentum that a star has will be conserved as it collapses and contracts under its own gravity, compounded by the additional angular momentum deposited by any infalling material. As such, while the simplified case of a Schwarzschild primary BH is useful as a testing ground to develop new theories and methods for EMRI waveform modelling, the final goal is always to extend any formalism to include Kerr BHs, as this is the astrophysically relevant case. This project will focus on a certain part of the Kerr parameter space, near-extremal Kerr BHs. To discuss this special case, a solid foundational understanding of the background Kerr spacetime is required, and much of the theory developed here will be used directly in following chapters, especially the discussion about orbits in Kerr spacetime, which is useful when discussing the trajectory of the secondary BH in the EMRI system.

2.2.1 The Kerr Metric

The angular momentum of a BH (J) is parametrised with the dimensionless Kerr spin parameter a in terms of the BH’s mass, $J = aM^2$, where we refer to a as the ‘spin’.

The metric used to describe the spacetime about a uniformly rotating object with mass M and spin a , is described by the Kerr metric [46], which in Boyer-Lindquist (BL) coordinates (t, r, θ, ϕ) [47], and with geometrised units ($G = c = 1$) is given as

$$ds^2 = g_{\mu\nu}^{\text{KERR}} dx^\mu dx^\nu = - \left(1 - \frac{2Mr}{\Sigma} \right) dt^2 - \frac{4Mar \sin^2 \theta}{\Sigma} dt d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \sin^2 \theta \left(r^2 + a^2 + \frac{2Ma^2 r \sin^2 \theta}{\Sigma} \right) d\phi^2, \quad (2.8)$$

where,

$$\Sigma = r^2 + a^2 \cos^2(\theta), \quad (2.9)$$

$$\Delta = r^2 - 2Mr + a^2 \quad (2.10)$$

This metric admits two Killing vectors $\varsigma = \partial/\partial t$ and $\vartheta = \partial/\partial \phi$ associated with time translations and rotations about the axis of rotation respectively. Each of these Killing vectors yield a *constant of motion*, the energy (E) and the z-component of the angular momentum (L_z) respectively, where the z-axis is aligned with the rotational axis of the BH. A third constant of motion was found in 1968 by Carter [48], aptly named the Carter constant $\mathcal{Q}$, but is of less significance to us, as it is related to the inclination of the orbit [49]. As shall be discussed in section 2.3, these three conserved quantities, along with the fourth conserved quantity of the rest mass of the test particle, fully define a geodesic trajectory, and so will be discussed in greater detail when introducing orbits in Kerr spacetime.

2.2.2 Singularities and Horizons of the Kerr Metric

In BL coordinates, the Kerr metric has singularities, where some terms of $g_{\mu\nu}^{\text{KERR}}$ will diverge. These points can be visually identified to be where $\Sigma = 0$ and $\Delta = 0$. We can classify these singularities as either *coordinate*, or *physical* (essential) singularities. A coordinate singularity is an artefact of the coordinate system that we are using (in this case BL coordinates), and will disappear if we move to a suitable coordinate system. A physical singularity however will remain singular no matter the coordinates. As such, we can test if a singularity is physical by computing some invariant scalar quantity. If the invariant blows up in one coordinate frame, it must blow up in every coordinate frame, such is the nature of a scalar invariant.

In both Schwarzschild and Kerr vacuum spacetimes, an adequate scalar to test these singularities is the Kretschmann scalar, $\mathcal{K} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, where $R_{\mu\nu\rho\sigma}$ is the Riemann tensor. The Kretschmann scalar is an interesting quantity in itself, packaging information about spacetime curvature around a BH, but the vital property of the Kretschmann scalar in this discussion is simply that it obeys scalar transformation laws, and thus is invariant under coordinate transformations. We will use the result of Ref. [50], in which the author calculates the Kretschmann scalar for a rotating and charged BH, but we can neglect the charge, which simplifies the result to

$$\mathcal{K} = \frac{48M^2 [r^6 - 15a^2r^4 \cos^2(\theta) + 15a^4r^2 \cos^4(\theta) - a^6 \cos^6(\theta)]}{(r^2 + a^2 \cos^2(\theta))^6}. \quad (2.11)$$

It is clear that the curvature thus goes to infinity at $\Sigma = 0$ (remembering the definition of Σ from Eq. (2.9)), but not at $\Delta = 0$. This verifies that $\Sigma = 0$ is in fact a true, coordinate-invariant singularity of the Kerr metric.

Thus, we can conclude that the singularity in Kerr spacetime has two conditions:

$$\Sigma = 0 \implies \begin{cases} \text{i)} & r = 0 \\ \text{ii)} & \theta = \pi/2. \end{cases} \quad (2.12)$$

The remainder of this project will not require any further discussion about the physical singularity of Kerr spacetime at all, but instead will be more concerned with the coordinate singularity of BL coordinates, known as the *event horizon*.

$\Delta = 0$ is a quadratic equation in r and thus admits two solutions,

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}, \quad (2.13)$$

with $r_+ \geq r_-$. The outer solution (r_+) is known as the event horizon, and the inner solution (r_-) is known as the Cauchy horizon [51]. The event horizon (r_+) thus determines the null hypersurface beyond which we have no causal contact with a particle which has crossed it, hence the colloquial name of ‘*the point of no return*’.

The event horizon will be revisited when discussing near-extremal Kerr BHs in section 2.5, as the next important hyper-surface to be introduced in the following section, the separatrix, coincides with the event horizon in the extremal limit (in BL coordinates).

2.3 Trajectories About a Kerr Black Hole

To discuss the trajectory of an EMRI, an understanding of timelike geodesics in Kerr spacetime is required. We will begin by discussing the generic form of the equations of motion for a timelike particle, before specialising to the simplified case of eccentric orbits in the equatorial plane. While the trajectory of an EMRI is expected to have inclination as well as eccentricity, i.e. fully generic, we leave the inclusion of inclination for future works, as here we are primarily focussed on quantifying the effects of eccentricity on the precision of the spin measurement that EMRIs can provide. This question could also be asked for inclination, but there are additional modelling difficulties that would make building a fully generic waveform model beyond the scope of this project. Thus, we restrict ourselves to the equatorial plane.

2.3.1 Eccentric Orbits in Kerr Spacetime

For a rotating black hole, we define the equatorial plane to be the plane that is tangent to the z -axis of our coordinate system, which is always aligned with the rotational axis of the Kerr BH. If a particle orbits in the equatorial plane, the following simplifications can be made,

$$\theta = \frac{\pi}{2} \text{ \& } \dot{\theta} = 0 \quad (2.14)$$

$$\mathcal{Q} = 0, \quad (2.15)$$

where θ is the BL coordinate describing the inclination relative to the equatorial plane, and $\mathcal{Q}$ is the Carter constant as discussed. The geodesic equations that describe the motion of a timelike particle in the equatorial plane are,

$$\left(\frac{dr}{d\lambda}\right)^2 = T^2 - \Delta [r^2 + (L_z - aE)^2], \quad (2.16)$$

$$\frac{d\phi}{d\lambda} = \frac{aT}{\Delta} - (aE - L_z), \quad (2.17)$$

$$\frac{dt}{d\lambda} = \frac{(r^2 + a^2)T}{\Delta} - a(aE - L_z), \quad (2.18)$$

where as before $T(r) = E(r^2 + a^2) - aL_z$, and also as before, the geodesic motion of a body is fully determined by the conserved quantities $\vec{P} = \{E, L_z, \mathcal{Q} = 0\}$.

There are two fundamental frequencies for an eccentric equatorial orbit,

$$\frac{1}{M}\hat{\Omega}_\phi = \frac{d\Phi_\phi}{dt}, \quad \frac{1}{M}\hat{\Omega}_r = \frac{d\Phi_r}{dt}, \quad (2.19)$$

where the hat indicates that they are dimensionless. The first of these measures how quickly an orbit around the azimuthal axis is completed, and the second measures how fast a timelike particle moves between the two radial turning points of the orbit. Ref. [52] derives analytical expressions for these frequencies.

It is possible, and indeed useful, to use an alternate set of quasi-Keplerian constants of motion, such as those discussed in depth in Refs. [53, 54]; the semi-latus rectum p , eccentricity e and inclination parameter $x_I = \cos(I)$. For bound orbits, there is a one-to-one relation between the constants of motion $\vec{P} = \{E, L_z, Q\}$ and $\vec{\mathcal{P}} = \{p, e, x_I\}$, where

$$r(\psi) = \frac{p}{1 + e \cos(\psi)}, \quad x_I = \cos(\theta_{\text{inc}}), \quad e = \frac{r_a - r_p}{r_a + r_p} \quad (2.20)$$

where ψ is defined to be a monotonically increasing phase, related to the radial oscillations. θ_{inc} is an inclination parameter, where $\theta_{\text{inc}} = \pi/2$ is the equatorial plane, and thus for orbits in the equatorial plane, $x_I = 1$. The eccentricity parameter e is a dimensionless quantity which determines how far from circular an orbit is, with $e = 0$ being circular, and orbits with $e > 1$ being hyperbolic unbound orbits. Gravitational radiation produced by EMRIs on hyperbolic orbits is discussed in depth in Ref. [55]. Just as in the Keplerian case, the point of closest approach of the orbit is called periastron (r_p), and the point of farthest approach is called apastron (r_a). These are the turning points of the orbit, where $\dot{r} = 0$. They are related to the semi-latus rectum by

$$r_p = \frac{p}{1 + e}, \quad r_a = \frac{p}{1 - e}. \quad (2.21)$$

These turning points can be seen clearly in the above figure 2.3 as the lighter and darker tracks of the trajectory, in the plot on the left, with $r_p \simeq 3.57M$ and $r_a \simeq 8.33M$.

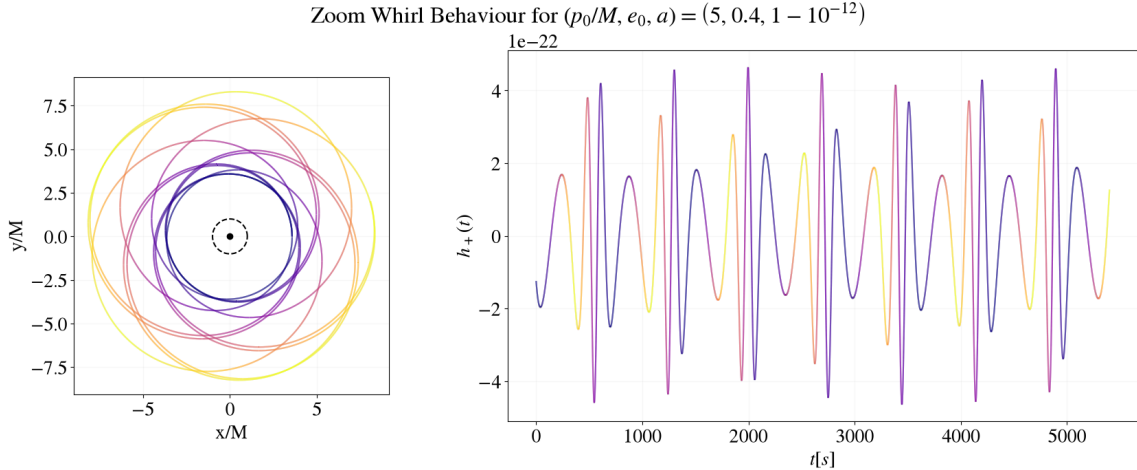


Figure 2.3: A snapshot of the trajectory and waveform for an EMRI with an orbital eccentricity of $e = 0.2$. The colour of the plots are coordinated, and changes based on radial distance to the event horizon, which is plotted as the dotted black circle.

Also displayed in the above figure, as a prelude as to what is to be discussed in later sections, is a gravitational waveform snapshot for the same trajectory. The “zoom-whirl”

waveform behaviour discussed in Ref. [56] can be seen as the amplitude of the waveform increases as the orbiting body moves closer to r_p , and decreases significantly as the secondary approaches r_a .

Now we return to our discussion of the turning points of radial motion. A turning point of the particles motion is characterised by points where $dr/d\lambda = 0$, or

$$E(r^2 + a^2) - aL_z - [r^2 - 2Mr + a^2][r^2 + (L_z^2 - aE)^2] = 0, \quad (2.22)$$

which after introducing the quantity $x = L_z - aE$, and a bit of manipulation, can be reduced to a cubic equation in r ,

$$r^3(E^2 - 1) + r^2(2M) - r(2aEx + x^2 + a^2) + 2Mx^2 = 0, \quad (2.23)$$

where in our algebra we cancelled a common term of r , which is fine as we won't be considering $r = 0$. Being a cubic equation in r , we can rewrite this equation in terms of its solutions

$$(E^2 - 1)(r - r_p)(r - r_a)(r - r_3) = 0. \quad (2.24)$$

Expanding this equation out, and comparing to equation (2.23) whilst using the expressions for r_a and r_p in terms of (p, e) allows us to solve for E , r_3 , and find an expression for $x(p, e)$,

$$E^2 = 1 - \left(\frac{M}{p}\right) (1 - e^2) \left(1 - \frac{x^2}{p^2}(1 - e^2)\right), \quad (2.25)$$

$$r_3 = \frac{2M(1 - e^2)x^2}{p^2(1 - E^2)}, \quad (2.26)$$

$$x^2 = \frac{A(p, e) \mp B(p, e)}{C(p, e)}, \quad (2.27)$$

where the functions A, B and C are given in Ref. [56], but the exact expressions are not important here. Through these equations, an expression for $E(p, e)$, as well as $L_z(p, e)$ (through the expression for $x^2(p, e)$) has been acquired. This gives a one-to-one relation between the conserved quantities E and L_z , and the quasi-Keplerian coordinates p and e .

We have explained the physical meaning of the two physical turning points of a particle's motion in an eccentric orbit, r_p and r_a , but what is this r_3 , the third solution to equation (2.23)? The point where $r_3 = r_p$ defines the innermost stable orbit (ISO), or the *separatrix* [53, 56], which in the Schwarzschild $a = 0$ case, reduces to the innermost stable *circular* orbit (ISCO),

$$p_{\text{isco}} = 6 + 2e. \quad (2.28)$$

Stable bound orbits only exist for $r_p > r_3$, and any timelike particle travelling on a trajectory that crosses beyond this point will plunge past the horizon of the BH. The separatrix is thus the separation between two key phases of an EMRI's lifetime; the inspiral, and the

plunge. A key restriction to the work done in the project is this fact, and a full inspiral-to-plunge model would be the ideal model, but is beyond the scope of this project. The problem of the inspiral-to-plunge regime is considered in sources such as Refs. [19, 57, 58].

It has been shown in Ref. [53], that the location of the separatrix is given by the roots of a 12th order polynomial in p , which depends on the quasi-Keplerian parameters. A numerical root-finding code can be implemented to efficiently find this root, and thus the separatrix. Such a code is part of the BHPT, and FEW.

It is worth taking some time to gain some insight in to how the parameters a and e affect the location of the separatrix, as the dependence of the separatrix on these parameters will arise repeatedly in our analysis. As can be seen in the figure 2.4 below, eccentricity has the effect of *increasing* the separatrix. In contrast, there is a strong inverse dependence of the separatrix on the spin parameter, which overrides the effect of the eccentricity in the extremal ($a \rightarrow 1$) limit, where the separatrix coincides with the event horizon at $r_+ = 1$.

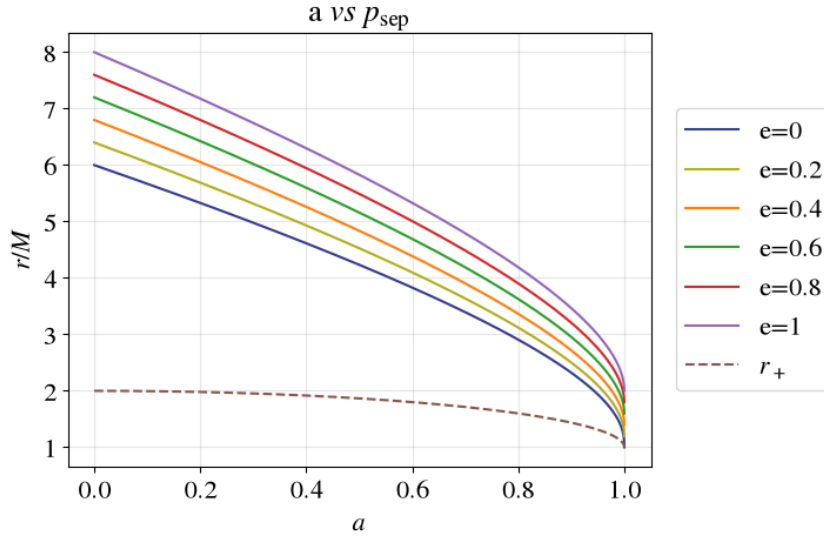


Figure 2.4: Plotted in the coloured lines are the values of the separatrix, p_{sep} for different values of eccentricity against the primary BH spin parameter a . The dotted line depicts the event horizon r_+ , also plotted against the same a values. Only prograde (positive spin) values are considered. The value of the separatrix at each point was calculated using FEW’s `get_separatrix` function.

The key fact here is that when working in BL coordinates, in the extremal limit the separatrix coincides with the event horizon, and thus we gain access to the *near-horizon regime*. The extremal limit is where we shall spend the rest of this dissertation in, and in section 2.5, we will explore how this near-horizon regime, and how EMRIs behave in this region of the parameter space.

2.4 Modelling of Extreme Mass Ratio Inspirals: Trajectories & Waveforms

The extreme mass ratio of the two bodies in an EMRI system allows for the application of black hole perturbation theory (BHPT), and gravitational self-force (GSF). In these formalisms the small secondary CO can be treated as a perturbation to the background spacetime of the primary MBH, and the perturbation ‘back-reacts’ on the secondary, resulting in what is known as a gravitational self-force [59, 60], or *radiation reaction*. It is this self-force that drives the inspiral. This section outlines the main tools used in GSF and BHPT to model the inspiral of the secondary BH, and ultimately calculate the GW strain detectable by LISA:

$$h(t) = h_+(t) - ih_\times(t). \quad (2.29)$$

2.4.1 Changing Constants of Motion?

The first step to calculating a GW waveform for an EMRI is to understand the trajectory of the inspiral. In the previous section, the theory required to understand how a timelike particle moves in the equatorial plane of a rotating BH was introduced. There, in the test-mass limit, two conserved quantities — E and L_z — fully determined the trajectory of the particle. The secondary CO in the EMRI system travels on a trajectory that is very close to that of a geodesic, but slowly loses energy and angular momentum. The rate at which the conserved quantities change is related to the GW flux via,

$$\begin{aligned} \dot{E} &= -\dot{E}^{\text{GW}}, \\ \dot{L}_z &= -\dot{L}_z^{\text{GW}}, \end{aligned} \quad (2.30)$$

where the negative signs are due to the fact that the secondary CO *loses* orbital energy and angular momentum in the form of gravitational radiation. Expressions for the fluxes of the quasi-Keplerian parameters $\dot{p}$ and $\dot{e}$ can then be found by recalling the equations (2.25) and (2.27), from which we can obtain an expression for $L_z(p, e)$. Then, through a chain rule expansion, $\dot{p}$ and $\dot{e}$ may be determined via,

$$\begin{bmatrix} \dot{E} \\ \dot{L}_z \end{bmatrix} = \begin{bmatrix} \frac{\partial E}{\partial p} & \frac{\partial E}{\partial e} \\ \frac{\partial L_z}{\partial p} & \frac{\partial L_z}{\partial e} \end{bmatrix} \begin{bmatrix} \dot{p} \\ \dot{e} \end{bmatrix} \implies \begin{bmatrix} \dot{p} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} \frac{\partial E}{\partial p} & \frac{\partial E}{\partial e} \\ \frac{\partial L_z}{\partial p} & \frac{\partial L_z}{\partial e} \end{bmatrix}^{-1} \begin{bmatrix} \dot{E} \\ \dot{L}_z \end{bmatrix}, \quad (2.31)$$

and thus the trajectory in quasi-Keplerian parameters can be solved for. Note however that the Jacobian matrix above becomes singular at the separatrix, and thus an alternative method is required when crossing the separatrix, but this is beyond the scope of this project. References [57, 58] discuss this problem in more depth in the eccentric and quasi-circular-near-extremal cases respectively. The components of the inverse Jacobian in the above equation can be obtained analytically.

All that is left to complete the right hand side of Eq. (2.31) are the fluxes, $\dot{E}$ and $\dot{L}_z$. There are many methods of doing so, each having its own advantages and disadvantages. Analytical methods such as post-Newtonian (PN) theory can calculate the flux quickly [61, 62]. The drawback is that PN expansions are not accurate in the strong-field regime of the primary BH. PN theory is thus not suitable for our requirements for this project, where the near-horizon regime is where the majority of the near-extremal phenomena will arise (to be discussed in section 2.5). Numerical relativity (NR), where the full Einstein equations are solved numerically, is well suited for systems with comparable mass (as seen in figure 2.5), where the timescale of each orbital cycle is approximately on the same order of the inspiral timescale. NR is not viable in the extreme mass ratio case due to the massive computational cost brought about by the long inspiral timescale [63, 64]. This leaves BHPT, which as can be seen in the figure below, perfectly covers our region of mass ratio - orbital separation parameter space.

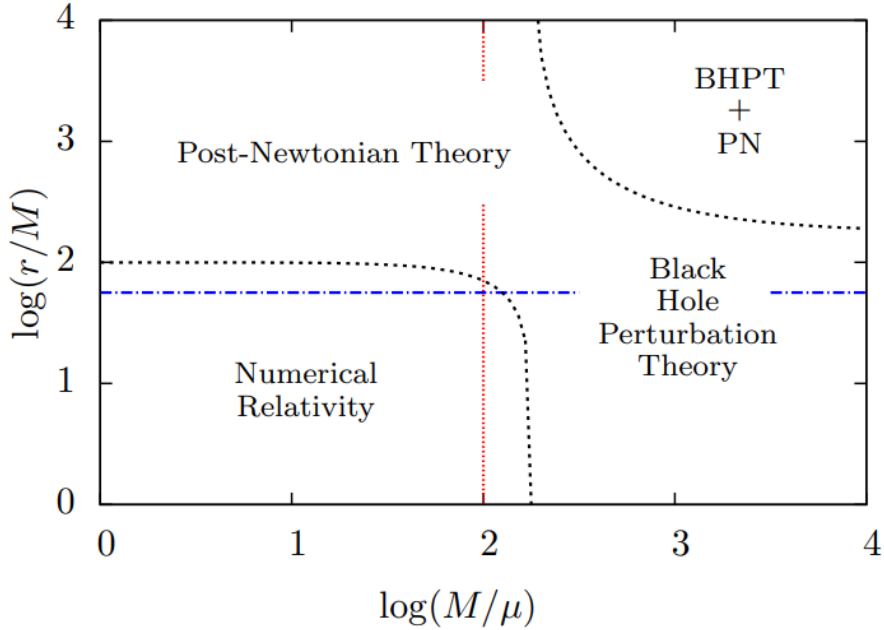


Figure 2.5: From Ref. [62], illustrating the regions of mass ratio - orbital separation parameter space where different methods for calculating GW fluxes are used.

The fluxes $\dot{E}$ and $\dot{L}_z$ can numerically be solved for using BHPT by utilising software packages such as PYBHPT [65], and converted to quasi-Keplerian fluxes $\dot{p}, \dot{e}$ using Eq. (2.31). After choosing a grid in parameter space on which to calculate the fluxes, as well as initial conditions of $\{a, p_0 > p_{sep}, e_0, x_I = 1\}$, Eq. (2.31) can be numerically integrated to solve for $\{p(t), e(t)\}$. For equatorial inspirals, $x_I(t)$ will not change, and will always be $x_I = 1$, simplifying the ODE system.

With the trajectory solved for, one can easily find the fundamental frequencies Ω_ϕ, Ω_r , using functions such as `get_fundamental_frequencies` from FEW [24]. Packages that

compute the gravitational fluxes also have the capability of computing the gravitational amplitudes which constitute the waveform, as the fluxes and amplitudes are closely related [56]. Indeed, we shall see in the following section 2.4.2 that the fluxes in energy and angular momentum are constructed from the *Teukolsky amplitudes*.

2.4.2 The Adiabatic Approximation and The Teukolsky Formalism

GSF theory begins with a perturbative expansion about any background metric, as outlined in Refs. [59, 60], in orders of some small parameter ϵ . In the context of EMRI WF modelling, the background metric is typically the Kerr metric of Eq. (2.8), and the small parameter is chosen to be the mass ratio of the EMRI,

$$g_{\mu\nu} = g_{\mu\nu}^{\text{kerr}} + \epsilon^1 h_{\mu\nu}^{(1)} + \epsilon^2 h_{\mu\nu}^{(2)} + \mathcal{O}(\epsilon^3). \quad (2.32)$$

In addition to being able to separate the behaviour of the motion of the perturbing body in terms of the mass ratio, the problem of modelling EMRIs can be further aided by utilising the two vastly different timescales of the problem, an approach known as the two-timescale approach [21, 66]. These time scales are the rapid orbital timescale T_{orbital} , and the inspiral timescale T_{inspiral} . These two timescales each scale with the masses involved in the EMRI system, as $T_{\text{orbital}} \sim M$ and $T_{\text{inspiral}} \sim M/\epsilon$. With these two time scales, the *two timescale* approach can be utilised. This approach involves the action-angle formulation of Ref. [66] where we introduce $\Phi_A = (\Phi_t, \Phi_r, \Phi_\theta, \Phi_\phi)$ which are a set of action angles in BL coordinates, and $J_A = (J_t, J_r, J_\theta, J_\phi)$ are actions associated with the angles [22, 66]. Using these, two equations of motion can be found,

$$\frac{d\Phi_A}{dt} = \Omega_A^{(0)}(J_B) + \epsilon \Omega_A^{(1)}(J_B) + \mathcal{O}(\epsilon^2), \quad (2.33)$$

$$\frac{dJ_A}{dt} = \epsilon \mathcal{F}_A^{(1)}(J_B) + \epsilon^2 \mathcal{F}_A^{(2)}(J_B) + \mathcal{O}(\epsilon^3). \quad (2.34)$$

where the first equation of motion varies on the fast timescale T_{orbital} , and the second is the slowly varying evolution of the orbital parameters over the entire inspiral. The general terms $\Omega_A^{(n)}$ and $\mathcal{F}_A^{(n)}$ are described in depth in sources such as Refs. [59, 66], but for our purposes we only require the first terms of each, $\Omega_A^{(0)}$ and $\mathcal{F}_A^{(1)}$.

Keeping only the first term in the expansion (2.33), and including only the orbit-averaged dissipative part of the first order forcing function $\mathcal{F}_A^{(0)}$ in the expansion in Eqs. (2.34) (as described in Refs. [19, 22, 65]), is known as the *adiabatic approximation*. After this, including an additional n terms in the equations of motion is also known as the n th post-adiabatic order, or nPA .

It is well-known that for accurate analysis of LISA data, and unbiased parameter estimation [21, 44, 67], 1PA waveforms are both necessary and sufficient. In this project however, only adiabatic order (0PA) terms are used to calculate the GW fluxes and amplitudes. This approximation is a limitation, not a choice. 1PA GSF theory is still under rapid

development, and while results are available for use in gravitational waveform models in limited cases, these cases do not include that of an eccentric EMRI about a near-extremal Kerr BH. Despite this, it is expected that 1PA corrections would have a negligible effect on the analysis performed in this work.

While 1PA is required for detection with LISA, developing a WF model with fluxes at 0PA is still well worthwhile. All important physical phenomena are captured at adiabatic order, so for the purposes of this project, 1PA is unnecessary. The modular framework of FEW allows for modifications to the flux and amplitude data to be easily made. As a result, once 1PA flux and amplitude data are available, they can be easily incorporated into the already built waveform models, i.e. the task of building a WF model in FEW remains the same, be it for 0PA or 1PA data [23].

At adiabatic order, methods from first order perturbation theory, such as the *Teukolsky formalism* for perturbations of Kerr spacetime, can be used to calculate the GW amplitudes and the fluxes of energy and angular momentum [68].

While the main goal of GSF theory is to reconstruct the metric perturbations $h_{\mu\nu}^{(n)}$, the Teukolsky formalism shifts the focus instead to the perturbations to Weyl curvature, which can be extracted via the Weyl curvature scalar, ψ_4 . This scalar encodes all the information needed to construct an adiabatic gravitational waveform as [56, 22]

$$\psi_4 = \frac{1}{2} \frac{d^2}{dt^2} (h_+ - ih_\times) \quad \text{as } r \rightarrow \infty. \quad (2.35)$$

Teukolsky showed in Ref. [68] that ψ_4 obeys a perturbative equation, and has separable solutions of the form,

$$\psi_4 = \frac{1}{(1 - ia \cos(\theta))^4} \int_{-\infty}^{\infty} d\omega \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} R_{\ell m}(r, \omega) S_{\ell m}(\theta, a\omega) e^{i(m\phi - \omega t)}, \quad (2.36)$$

where now $R_{\ell m}(r, \omega)$ and $S_{\ell m}(\theta, a\omega)$ satisfy ordinary differential equations, and the index ℓ corresponds to the multipole moment of the GW. Details of solving these equations can be found in Refs. [22, 65, 56]. The solutions to the radial equation yield the *Teukolsky amplitudes* — $Z_{\ell m \omega}^{\infty, H}$ — which describe the asymptotic behaviour of the radial solution — $R_{\ell m}(r, \omega)$ — as $r \rightarrow \infty$, and as $r \rightarrow r_+$. These amplitudes can be further broken into specific frequency-mode amplitudes

$$Z_{\ell m \omega}^{\infty, H} = \sum_{k=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} Z_{\ell m k n}^{\infty, H} \delta(\omega - \omega_{mkn}), \quad (2.37)$$

where $\omega_{mkn} = m\Omega_\phi + k\Omega_\theta + n\Omega_r$ is comprised of the fundamental frequencies of motion. Note that in the equatorial case, only modes with $k = 0$ will contribute to the above equation [24], and therefore the subscript k will be neglected, with the implication being that $k = 0$. These solutions are then used to reconstruct ψ_4 at $r \rightarrow \infty$ as,

$$\psi_4 = \frac{1}{r} \sum_{\ell m n} Z_{\ell m n}^{\infty} S_{\ell m}(\theta, a\omega_{mn}) e^{i(m\phi - \omega_{mn}t)}, \quad (2.38)$$

and thus Eq. (2.35) can be used to yield a *snapshot* waveform valid on a single geodesic,

$$h(t) = h_+(t) - ih_\times(t) = \sum_{\ell mn} A_{\ell mn} S_{\ell m}(\theta, a\omega_{mn}) e^{i(m\phi - \omega_{mn}t)}, \quad (2.39)$$

where the amplitude of the waveform is determined by,

$$A_{\ell mn} = -\frac{2Z_{\ell mn}^\infty}{\omega_{mn}^2}. \quad (2.40)$$

The functions $S_{\ell m}(\theta, a\omega_{mn})$ are the spin-weighted spheroidal harmonics, which will have an effect on the amplitude of the waveform depending on the viewing angle of the detector. They are discussed in greater detail in Ref. [69], however, they are computationally expensive to compute, and their computation can be circumnavigated by means of an expansion of spherical harmonics, which are more efficient to compute [23, 70].

The Teukolsky amplitudes of Eq. (2.37) are also used to calculate the dissipative effects of gravitational radiation on the energy and angular momentum of the perturbing body, in the context of this project, the secondary compact body of the EMRI system. While only the Teukolsky amplitudes at infinity comprise the waveform amplitudes, the total Teukolsky amplitudes at infinity *and* at the horizon are required to describe the change in the orbital dynamics of the perturbing mass. These fluxes are found as a sum of Teukolsky amplitudes,

$$\begin{aligned} \left(\frac{dE}{dt}\right)^{\infty, H} &= \sum_{\ell mn} \frac{\alpha_{\ell mn}^{\infty, H} |Z_{\ell mn}^{\infty, H}|^2}{4\pi\omega_{mn}^2}, \\ \left(\frac{dL_z}{dt}\right)^{\infty, H} &= \sum_{\ell mn} \frac{m\alpha_{\ell mn}^{\infty, H} |Z_{\ell mn}^{\infty, H}|^2}{4\pi\omega_{mn}^3}, \end{aligned} \quad (2.41)$$

where $\alpha_{\ell mn}^{\infty, H}$ is a constant given in Refs. [65, 22]. The total GW energy flux is then simply given by the sum,

$$\begin{aligned} \dot{E}^{GW} &= \dot{E}^\infty + \dot{E}^H, \\ \dot{L}_z^{GW} &= \dot{L}_z^\infty + \dot{L}_z^H, \end{aligned} \quad (2.42)$$

which can each be used in Eqs. (2.30). Thus, to solve for the waveform amplitudes, and the energy and angular momentum fluxes of the system, the Teukolsky amplitudes will be solved for numerically using PYBHPT [65].

Note that Eq. (2.39) does not fully capture how the waveform evolves with time over the course of the entire inspiral, but rather provides a snapshot of the waveform on a particular geodesic. However, due to the loss of energy and angular momentum determined by Eqs. (2.42), the secondary BH adiabatically evolves from one geodesic to another, and thus the energy, angular momentum and the frequencies must be functions of time. To handle this, a time parameter that tracks the evolution of the inspiral according to a distant observer — t^i — must be introduced. Recalling Eq. (2.19), this expression may now be re-written as [65],

$$\Phi_A(t) = \int \Omega_A dt, \quad (2.43)$$

where $A = \{\phi, r\}$. Incorporating this phase allows the waveform from Eq. (2.39) to fully capture the effect of radiation reaction on the inspiral, to make an adiabatic waveform,

$$h = h_+ - ih_\times = \sum_{\ell mn} A_{\ell mn}(t^i) S_{\ell m}(\theta, a\omega_{mn}(t^i)) e^{i(m\phi - \Phi_{mn}(t^i))}, \quad (2.44)$$

which is a subtle, yet impactful change from Eq. (2.39), as without it, the waveform would de-phase on a time scale of $M/\sqrt{\epsilon} > T_{\text{inspiral}}$ [66, 22], the errors from which would be unsuitable for our purposes of using this adiabatic waveform for data analyses of EMRI waveforms about near-extremal MBHs.

2.4.3 FastEMRIWaveforms

The FastEMRIWaveforms Python package [23, 24, 70] is a codebase dedicated to the rapid generation of gravitational EMRI WFs for data analysis, and for use in matched filtering algorithms for LISA, and for Bayesian algorithms. The package interpolates pre-computed GW flux and amplitude data, and numerically integrates the trajectory of the EMRI, and computes the amplitudes of each WF mode. The FEW framework is designed to be highly modular, so it is as easy as possible to implement new physics such as 1PA data, or environmental effects.

A high-level overview into the most relevant stages of the waveform modelling process that is completed by FEW are outlined in the figure below. Note that the process of computing the fluxes using PYBHPT does not occur within FEW, but instead are precomputed on a grid in geodesic parameter space (a, p, e, x_I) , and this data grid is then fed into FEW to be interpolated. This is how waveforms can be computed so rapidly.

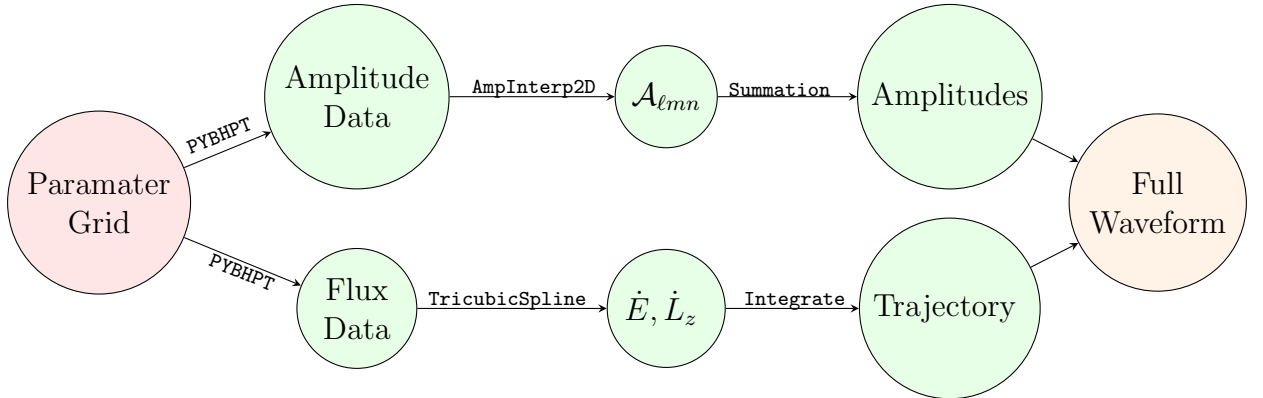


Figure 2.6: Overview of the milestones on the way to a complete waveform with FEW.

To solve for the adiabatic inspiral using the flux data, FEW integrates a total of 6 ODEs, all of which come from the adiabatic order of expansions (2.33) and (2.34). To better match results with NR, FEW opts to re-expand these equations in terms of the symmetric

mass ratio $\nu = M\mu/M_{\text{tot}}^2$, which is viable as $\epsilon = \nu + \mathcal{O}(\nu^2)$ [24, 71]. To adiabatic order, these equations are,

$$\frac{d\alpha}{dt} = \frac{\nu}{M} (f_{(0)}^\alpha(a, p, e, x_I) + \mathcal{O}(\nu)), \quad (2.45)$$

$$\frac{d\Phi_A}{dt} = \frac{1}{M} \hat{\Omega}_A, \quad (2.46)$$

where $\alpha = (p, e, x_I)$, and the forcing function $f_{(0)}^\alpha$, is found by combining the adiabatic order of expansion (2.34), and the inverse Jacobian of Eq. (2.31). Upon integrating these EOM FEW gains access to the full trajectory of the EMRI, and can then work on solving for Eq. (2.44) as efficiently as possible.

FEW bypasses the computation of the spin-weighted spheroidal harmonics entirely, by expanding in the spherical harmonics [70, 23],

$$S_{\ell mn}(\theta, \hat{\omega}_{mn}) e^{im\phi} = \sum_{j=\ell_{\min}}^{\infty} b_{\ell mn}^j(t) {}_{-2}Y_{jm}(\theta, \phi), \quad (2.47)$$

where the coefficients $b_{\ell mn}^j$ are given in [69]. This means that Eq. (2.44) can be re-written as,

$$h(t) = \sum_{\ell, m, n} \mathcal{A}_{\ell mn}(t) {}_{-2}Y_{\ell m}(\theta, \phi) e^{-\Phi_{mn}(t)}, \quad (2.48)$$

where now,

$$\mathcal{A}_{\ell mn} = \sum_{j=\ell_{\min}} b_{\ell mn}^j A_{jmn}, \quad (2.49)$$

are the amplitude quantities that FEW uses to construct the WFs, where A_{jmn} is the quantity from Eq. (2.40). With these amplitudes, the waveform can then be obtained. It is expressed in Ref. [24] as,

$$h(t) = \frac{\mu}{D} \sum_{\ell mn} \mathcal{A}_{\ell mn}(t) {}_{-2}Y_{\ell m}(\theta, \phi) e^{-i\Phi_{mn}(t)}, \quad (2.50)$$

where D is the luminosity distance to the source.

2.5 Near-Extremal Kerr Black Holes

Near-extremal BHs provide a fascinating glimpse into the realm of quantum gravity, and represent an active area of research in both mathematics and astrophysics. This section will introduce some of the reasons why near-extremal BHs are so interesting to physicists and mathematicians, and also briefly explore some mechanisms that may allow a near-extremal BH to form, surpassing the ‘‘Thorne limit’’. We will then discuss a simplified EMRI system, reduced to the case of a quasi-circular inspiral around a near-extremal BH. This analytical study will provide the foundation of the understanding of our results.

2.5.1 Physical Feasibility of Near-Extremal Kerr Black Holes

It seems fair to say that across the astrophysical community, rapidly rotating black holes, with spins as high as will be considered in this project, are not generally considered to be physically abundant systems. This is mostly due to the fact that most scenarios that involve increasing the angular momentum, or “spinning up”, a BH involve accretion of infalling material. In Ref. [26], Thorne derived the “*Thorne limit*”, which represents the maximum possible spin that a BH can reach via accretion of infalling material, which is balanced by a torque induced by radiation, which counteracts the spin-up effect of the infalling material resulting in a maximum possible spin of $a \simeq 0.998$. This, unlike the cosmic censorship conjecture, is not seen to be such a stringent constraint as alternate mechanisms for spinning up a BH past this limit have been shown to exist on paper, and may well exist in nature.

Not only is accretion limited in its effectiveness in spinning up a BH, but the authors of Ref. [72] utilise the same BHPT techniques outlined in the prior section to show that test particles on a quasi-circular inspiral will not spin up the BH that they are orbiting past a very similar limit of $a \leq 0.9979$.

Primordial BHs are especially interesting candidates for near-extremal BHs. These are BHs that form via the gravitational collapse of extreme over-densities in the early universe [73]. Ref. [28] directly shows how a primordial BH may be spun up through the emission of Hawking radiation. This suggests that the detection of a near-extremal BH could give hints into how primordial BHs could have formed.

In conclusion of this short discussion about the physical feasibility of near-extremal BHs, the evidence for them thus far in observations of BHs is lacking [74], and the suggested mechanisms for their formation are few. However, we opt to take the mindset that they are worth searching for with LISA, as the implications of their existence are so consequential to a wide range of fields in physics and mathematics.

2.5.2 EMRIs with a Near-Extremal Primary

The analytical example that is studied in this subsection will provide a foundation of the intuition that will be applied in our results. Here we study the *circular*, equatorial EMRI about a near-extremal BH, which has been studied in detail, both analytically and numerically, in sources such as Refs. [19, 29, 30, 75]. While this dissertation focuses primarily on *eccentric* equatorial EMRIs, the near-horizon-near-extremal effects studied in these works are greatly applicable. In section 2.3.1, we demonstrate that the critical value of p at which the sign of $\dot{e}$ turns positive is far closer to the separatrix than for more moderately rotating BHs, and thus we would expect EMRIs about a near-extremal BH to circularise more efficiently (as is demonstrated in section 4.1). Thus, the formalism developed here, which focuses on the near-horizon dynamics of the EMRI, is all the more applicable.

An integral part of EMRI dynamics about a near-extremal primary is a result of the separatrix approaching the event horizon in the extremal limit, as discussed in section 2.3.1. The behaviour of the gravitational radiation that is emitted by an orbiting body in the near-horizon regime of a near-extremal BH results in both the waveform amplitudes, and the inspiral dynamics being altered in such a way that the GW signal from one of these systems is unmistakable, and can serve as a “smoking gun” [29, 57], meaning that should a waveform of this shape be detected, it is almost certain that it came from a near-extremal system. Here, this smoking gun is derived analytically, following from Refs. [19, 29].

Look again at the solutions to $\Delta = 0$:

$$r_{\pm}/M = \tilde{r}_{\pm} = 1 \pm a\sqrt{1 - (a/M)^2} \quad (2.51)$$

where $\tilde{r}_+$ is the *dimensionless* event horizon. Now introduce some useful dimensionless parameters,

$$\epsilon = \sqrt{1 - a^2/M^2} \sim \lambda \quad (2.52)$$

$$x = \frac{r - r_+}{r_+} \sim \lambda^{2/3}, \quad (2.53)$$

which are a measure of how far the primary BH is from being extremal, and how far the secondary BH is from the horizon respectively, and λ is an order counting parameter. Then, the energy and frequency for a circular geodesic close to the horizon of a near-extremal BH, where $\lambda \ll 1$, can be written as [29],

$$E = \frac{\mu}{\sqrt{3}} \left(1 + \frac{2x_0}{3} \left(1 + \frac{\epsilon^2}{x_0^3} \right) \right), \quad (2.54)$$

$$\Omega_{\phi} = \frac{1}{2M} \left(1 - \frac{3x_0}{4} \right). \quad (2.55)$$

In Ref. [75], the authors utilise the Teukolsky formalism introduced in section 2.4.2, along with the near-extremal parameters x and ϵ , to derive the energy flux of a particle close to the ISCO. It is shown that at leading-order, the energy flux at infinity and at the horizon scale linearly with x , i.e. $\dot{E}^{\infty} = C^{\infty}x$, $\dot{E}^H = C^Hx$, where the constants C^H and C^{∞} can be found in equations (76) and (77) of Ref. [75] respectively. Thus the total energy flux (and power) of the GW can be expressed as,

$$\dot{E}^{\text{GW}} = (C^{\infty} + C^H)x. \quad (2.56)$$

To find how the inspiral of a particle losing this orbital energy evolves, we need an ODE for $r(t)$, or equivalently here $x(t)$, where t is time according to a distant observer. This can be found using Eqs. (2.54) and (2.56), and utilising the chain rule to find,

$$\frac{dx}{dt} = - \left(\frac{dE}{dx} \right)^{-1} \dot{E}^{\text{GW}} = \frac{x}{\tau} \left(\frac{1}{1 - 2\epsilon^2/x^3} \right), \quad (2.57)$$

where τ is the timescale,

$$\tau = \frac{2}{3\sqrt{3}} \frac{\mu}{C^{\infty} + C^H}. \quad (2.58)$$

This ODE has an exact analytical solution of

$$x(t) = X_0 e^{-t/\tau} g(t), \quad (2.59)$$

where $x(t=0) = X_0$ and,

$$g(t) = \exp\left(\frac{k}{3} + \frac{1}{3}\mathcal{W}(-ke^{3t/\tau-k})\right), \quad (2.60)$$

where $k = \frac{2\epsilon^2}{X_0^3}$, and where $\mathcal{W}$ is the product log function [76]. The function $g(t)$ will not have an impactful effect on the trajectory for earlier times, and so we can say,

$$x(t) \sim e^{-t/\tau}, \quad (2.61)$$

and thus using Eq. (2.56),

$$P_{\text{detector}} \sim e^{-t/\tau}, \quad (2.62)$$

meaning that the power of the GW that a detector observes *decreases exponentially* as the inspiral evolves into the near-horizon regime. This results in a characteristic dampening of the waveform amplitudes close to the horizon, and thus close to the separatrix of near-extremal BHs. This is the first smoke signal from the smoking gun.

The only two observables that a GW detector can measure are the amplitude and the frequency of a GW. The first component of the smoking gun is the exponential decrease of the amplitude, and the second component is the behaviour of the frequency. Note the horizon frequency of a BH

$$\Omega_H = \frac{1}{2Mr_+}. \quad (2.63)$$

In the extremal limit, $r_+ \rightarrow 1$, and thus the horizon frequency of an extremal BH is simply $1/2M$, which is exactly the prefactor of Eq. (2.55). The frequency that a GW detector would then detect is [29],

$$f_{\text{detector}} \sim 2 \times \frac{\Omega_H}{2\pi} \simeq \frac{1}{2\pi M}, \quad (2.64)$$

or in SI units

$$f_{\text{detector}} \simeq 3.2 \times 10^4 \left(\frac{M_\odot}{M}\right) \text{ Hz}. \quad (2.65)$$

Thus the second smoke signal of our smoking gun is that the orbital frequency of the EMRI saturates the horizon frequency of the primary, and the detector observes a WF with a frequency given by Eq. (2.65).

These “smoking gun” characteristics are what are expected for a quasi-circular inspiral. We will explore how eccentricity affects them.

2.6 Constraining BH spin with EMRIs

The fundamental task of LISA is to extract the parameters of a GW source as accurately as possible. Prior work has shown that in the case of a quasi-circular EMRI about a near-extremal Kerr primary BH, the measurement of the spin parameter of the primary BH has the possibility to be extracted with a precision of $\sim 3 - 4$ orders of magnitude higher than a non-extremal primary [30]. This analysis can be performed by calculating the Fisher information matrix. This section will describe the formalism behind the basic estimations that can be made about how well LISA will be able to constrain the parameters of an EMRI system.

Throughout this discussion, consider an eccentric EMRI GW source, with parameters $\boldsymbol{\theta}$,

$$\boldsymbol{\theta} = \{a, p_0, e_0, M, \mu, \Phi_{\phi,0}, \Phi_{r,0}\}, \quad (2.66)$$

where a subscript of 0 indicates an initial value. LISA's objective is to determine these parameters with as high-precision as possible. The LISA data stream consists of a superposition of GW signals, and some probabilistic noise. For simplicity, we will take there to be only a single GW signal, neglecting the remaining signals. This way we can analyse the impact of a signal source. With this approximation, the LISA data stream takes the form,

$$d(t) = h(t; \boldsymbol{\theta}) + n(t). \quad (2.67)$$

The power of the noise $n(t)$ can be quantified by the power spectral density (PSD), $S_n(f)$, which gives the power of the noise at a given frequency f . Ref. [18] gives an overview of the considerations involved when constructing the PSD. The PSD can be used to calculate the *noise weighted inner product*, which is given by Ref. [77] as,

$$(b|c) = 4\mathcal{R}e \int_0^\infty \frac{\hat{b}(f)\hat{c}^*(f)}{S_n(f)} df, \quad (2.68)$$

where b and c are some continuous data streams, and $\hat{b}$, $\hat{c}$ are their Fourier transforms. This noise weighted inner product allows for the calculation of several key properties of a GW signal. First, and most simple, is the signal-to-noise ratio (SNR) ρ [30, 29],

$$\rho^2 = (h|h), \quad (2.69)$$

$$= 4 \int_0^\infty \frac{|\hat{h}(f)|^2}{S_n(f)} df. \quad (2.70)$$

This quantity represents how loud the signal is compared to the background noise $n(t)$ and thus how difficult it will be to detect it.

The next useful piece of information that can be extracted with the noise weighted inner product is the Fisher information matrix Γ , with entries Γ_{ij} , which can be calculated as [30, 78]:

$$\Gamma_{ij} = \left(\frac{\partial h}{\partial \theta_i} \middle| \frac{\partial h}{\partial \theta_j} \right), \quad (2.71)$$

with $h \equiv h(t; \boldsymbol{\theta})$. The inverse of the Fisher matrix can be used to approximate the covariance matrix of the parameter precisions $\Delta\theta_i$ [30],

$$\text{Cov}(\Delta\theta_i, \Delta\theta_j) \approx (\Gamma^{-1})_{ij}. \quad (2.72)$$

The diagonal elements of this covariance matrix provide the variances of the parameters when correlations with all other parameters have also been considered, and thus the square roots of the diagonal elements give estimates of the 1σ standard deviation of the measurement of the parameters when other parameters have been taken into account [30, 19],

$$\theta_i^2 = (\Gamma^{-1})_{ii}, \quad (2.73)$$

where the double index does not imply any summation, but rather a single diagonal element.

As discussed, prior work in Ref. [30] has demonstrated that the spin parameter can be measured with very high-precision for a quasi-circular near-extremal EMRI, but no work since has conducted the same analysis with the inclusion of eccentricity, and thus also the initial radial phase parameter $\Phi_{r,0}$. An efficient waveform generator that includes the near-extremal regime of the parameter space allows for these prior results to be extended to include the effects of eccentricity.

These tools are defined for *continuous* signals, we must adapt them to accommodate signals that have been discretely sampled, as discussed in Ref. [19]. The noise weighted inner product, Eq. (2.68) can be well approximated by the discrete sum

$$(a|b) \simeq 4\Delta t \text{Re} \sum_{i=0}^{N/2+1} \frac{\hat{a}(f_i)\hat{b}(f_i)}{S_n(f_i)}, \quad (2.74)$$

which can also be used to compute the SNR (Eq. (2.70)).

The elements of the Fisher information matrix, Eq. (2.71) can be approximated with a five point stencil finite difference scheme, the same that is adopted in Ref. [30],

$$\frac{\partial f}{\partial x} = \frac{-f_{(2)} + 8f_{(1)} - 8f_{(-1)} + f_{(-2)}}{12\delta x} + \mathcal{O}(\delta x^4), \quad (2.75)$$

where $f_{(i)} = f(x + [i] \times \delta x)$ and $\delta x \ll 1$. This finite difference scheme was accurate enough to capture small changes in the waveform when the parameter the derivative is being taken with respect to does not have a large effect on the WF. Specifically, the initial radial phase parameter $\Phi_{r,0}$, which has the least effect on the waveform, and required a finite difference scheme with an error of $< \mathcal{O}(\delta x^2)$.

With these tools at hand, all that is required to extend the results of Ref. [30] to include orbital eccentricity, is a suitable waveform model.

Chapter 3

Model Implementation and Validation

With the relevant background theory and formalism discussed, we can now implement a waveform model within the FastEMRIWaveforms framework. This chapter discusses and justifies the decisions made when building the model. We then test the accuracy of the implemented trajectory and waveform generators and also highlight the model’s limitations.

There are two primary sources of systematic error when modelling EMRI waveforms with precomputed perturbation theory data. These are error induced by mode-sum truncation and interpolation error [79]. Due to this model being one that is in early development, the fluxes and amplitudes are calculated on a relatively sparse grid. As a result the primary source of error prevalent here is expected to be *interpolation error*. As such, to test the accuracy of the model built for this project, we compare the interpolated flux and amplitude functions to their counterparts in two limiting cases:

- The lower-spin end parameter space on which the near-extremal model is developed overlaps with the high-spin end of the existing FEW model, namely the spins overlap on $a \in [0.99, 0.999]$. This FEW model has itself been compared to error-free adiabatic waveforms, achieving mismatches of just $\sim 10^{-5}$ [24].
- Flux data for quasi-circular inspirals, provided by the primary supervisor of this project, allows for the comparison of the interpolated flux functions in the circular case.

Upon testing the accuracy of interpolated values of the fluxes and amplitudes, we can compare the output of the model against FEW’s eccentric model.

3.1 Parameter Grid

The FastEMRIWaveforms package leverages pre-processed data which can be interpolated to obtain smooth functions for the flux and amplitudes across the parameter space.

This pre-processing involves the calculation of the waveform amplitudes from Eq. (2.40), and GW fluxes from Eq. (2.41). These quantities are computed on a 3-dimensional grid of parameters (a, p, e) . To reduce interpolation error, the density distribution of the points of this grid should reflect the regions of parameter space where the flux and amplitudes are expected to vary the most.

The interpolation takes place over interpolation coordinates (u, w, z) , where each interpolation coordinate varies uniformly between the values of 0 and 1. The coordinates u, w, z form an evenly sampled rectilinear grid, with (N_u, N_w, N_z) points in the (u, w, z) directions respectively. The mappings between (u, w, z) and (a, p, e) are what determines the density distribution of points in the parameter space. Here we define the bounds of our model,

$$\Delta p_{\min} \leq p - p_{\text{sep}} \leq \Delta p_{\max} \quad \text{where:} \quad \Delta p_{\min} = 10^{-2}, \quad \Delta p_{\max} = 10, \quad (3.1)$$

$$0 \leq e \leq e_{\max} \quad \text{where:} \quad e_{\max} = 0.4, \quad (3.2)$$

$$a_{\min} \leq a \leq a_{\max} \quad \text{where:} \quad a_{\min} = 1 - 10^{-2}, \quad a_{\max} = 1 - 10^{-12}. \quad (3.3)$$

In the near-extremal limit, the amplitudes and fluxes are expected to vary rapidly close to the separatrix (p_{sep}), as they exhibit the near-horizon dampening described in section 2.5. To accurately capture this behaviour in our model, we adopt the following mappings,

$$u = \frac{\log_{10}(p - p_{\text{sep}}[a, p, e, x_I]) - \log_{10}(\Delta p_{\min})}{\log_{10}(\Delta p_{\max}) - \log_{10}(\Delta p_{\min})}, \quad (3.4)$$

$$w = \frac{e}{e_{\max}}, \quad (3.5)$$

$$z = \frac{\log_{10}(1 - a) - \log_{10}(1 - a_{\min})}{\log_{10}(1 - a_{\max}) - \log_{10}(1 - a_{\min})}. \quad (3.6)$$

These mappings put a heavy bias in the density of grid points in p towards the separatrix, and in a towards the near-extremal limit. We opt for a linear mapping in e to evenly sample the eccentricities. These mappings are used for both the flux and the amplitude data.

3.2 Building a Fast EMRI Waveform Model

Once the grid has been established, it can be integrated into the FastEMRIWaveforms framework. New inspiral and waveform classes can then be built in FEW. Figure 2.6 shows a high-level overview of how these classes fit into the overall architecture of FEW. The `KerrEccEq` model can be readily modified to accept the new parameter space defined by the grids. This is straightforward, but the entire codebase also has $\sim 30,000$ lines of code (not including example Jupyter notebook files). Understanding which of these 30,000 lines of code needs to be changed can quickly become a non-trivial task. A step-by-step overview of what needs to be altered in the FEW code to implement a new model on a new region of parameter space is given in Appendix B.

3.3 Flux Validation

The gravitational fluxes $\dot{E}$ and $\dot{L}_z$ are computed on the grid described by equations (3.4)-(3.6), with $(N_u, N_w, N_z) = (14, 9, 9)$. Additional points in u (and thus p) are sampled as the near-horizon behaviour proved to be difficult to accurately model without them.

As eccentricity increases, the number of n modes required for convergence of Eq. (2.37) increases. This requirement for large amounts of n modes is mirrored by the ℓ modes close to the separatrix. As the separatrix approaches the event horizon in the near-extremal limit, a large number of ℓ modes are required to adequately resolve the near-horizon behaviour. These computational challenges are shown below, where the wall time for a selection of flux data points is plotted.

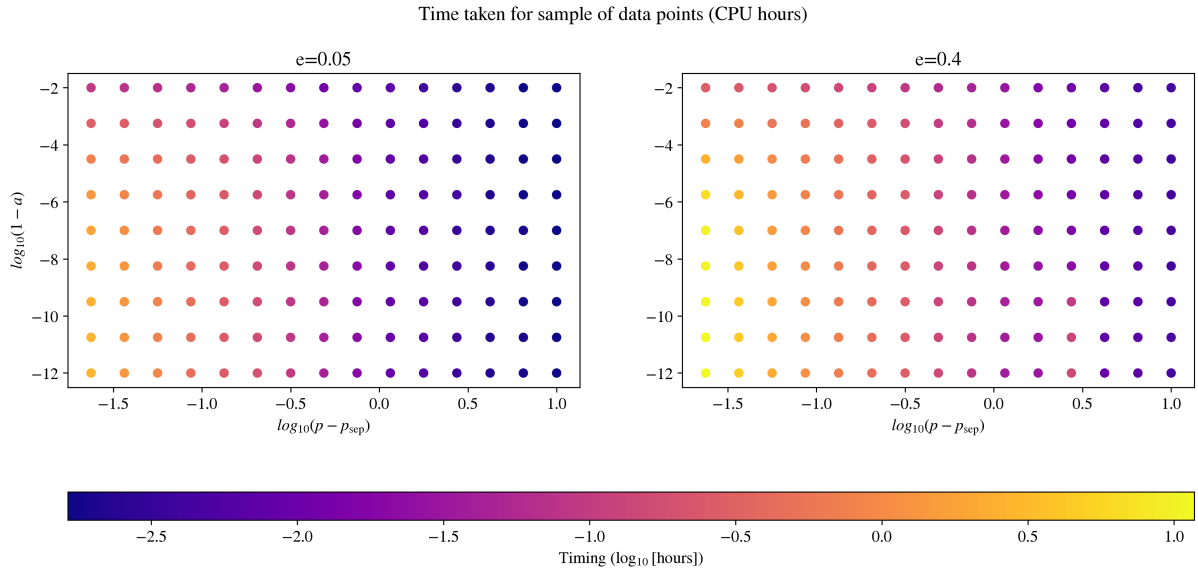


Figure 3.1: Shown is the wall time in CPU hours for the computation of the fluxes $\dot{p}$ and $\dot{e}$ on a portion of the parameter grid, for $e = 0.05$ and $e = 0.4$. The time taken to evaluate the fluxes close to the separatrix of an MBH with a spin parameter of $a = 1 - 10^{-12}$, where the secondary CO is on an orbit with eccentricity of $e = 0.4$, is ~ 10 CPU hours.

From the above figure, it is clear that the computation time of the fluxes has a very high dependence on the semi-latus rectum p . Flux values calculated at data points that are close to the separatrix, with high eccentricity and spin, are highly computationally expensive. This informs our choice of $\Delta p = 10^{-2}$, as computing the fluxes any closer to the separatrix would be too computationally expensive on the time frame of this project.

The Python package `multispline` is used to generate a tri-cubic spline over the interpolation coordinates (u, w, z) . This can be used with the mappings from Eqs. (3.4)-(3.6) to calculate the interpolated flux at any point in the parameter space defined by Eqs. (3.1)-(3.3). These functions shall henceforth be known as $\dot{E}^{\text{NEX}}$ and $\dot{L}_z^{\text{NEX}}$ (where NEX is shorthand for near-extremal). Eq. (2.31) can be implemented using FEW's `ELdot_to_PEdot_Jacobian` function to obtain equivalent expressions for $\dot{p}^{\text{NEX}}$ and $\dot{e}^{\text{NEX}}$,

where the Jacobian can be applied before or after interpolation. Before the tri-cubic spline is generated, the leading-order PN behaviour is divided out of the fluxes, this greatly reduces interpolation error further from the separatrix where the first leading-order PN term captures the fluctuating behaviour. The same PN term is multiplied back in *after* interpolation has been completed, to regain the flux values.

To compare interpolated fluxes, we compute the relative differences between the models. We stress that this is not the relative *error*, as this would imply that the model we are comparing against has no error of its own. We define

$$\delta(\dot{\alpha}^A, \dot{\beta}^B) = \log_{10} \left| 1 - \frac{\dot{\alpha}^A}{\dot{\beta}^B} \right|, \quad (3.7)$$

where $\alpha = (p, e, E, L_z)$, and $A = (\text{NEX}, \text{FEW}, \text{CIRC})$ indicates what models/data sets are being compared. We use the superscript FEW to indicate the KerrEccEq model in FEW, and CIRC to indicate the zero-eccentricity (circular) data.

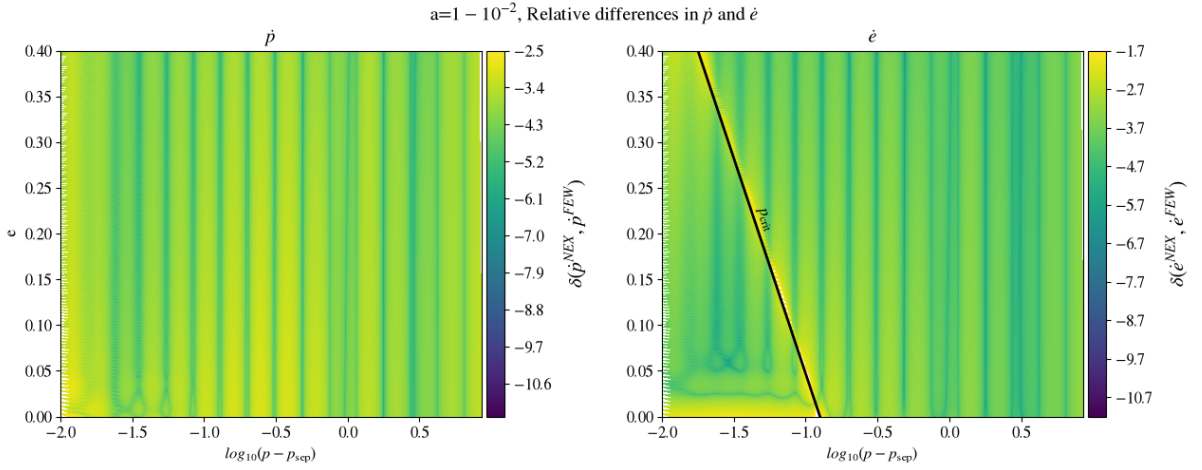


Figure 3.2: Displayed are the relative differences in the interpolated functions of $\dot{p}$ (left) and $\dot{e}$ (right) of the FEW and NEX models. The spin parameter of the primary MBH is fixed at $a = 1 - 10^{-2}$. The line p_{crit} indicates the value of p along which $\dot{e}^{\text{FEW}} = 0$, and thus introduces artificial increases in the relative differences as the denominator of Eq. (3.7) approaches 0. This value of p_{crit} will be discussed when discussing the evolution of $e(t)$ along the trajectory.

The relative differences above give confidence that the interpolation functions provide good agreement in the lower spin limit with FEW. The dark vertical bars where the relative difference is on the order of $\delta \lesssim 10^{-5}$, show clearly where the data points directly computed by PYBHPT lie. The small value of the relative differences at these points confirm that interpolation error is the dominant error in the model.

The plots in figure 3.3 show the relative differences between the interpolated fluxes of the near-extremal model and the same interpolated fluxes that were computed independently in the circular case. The highest differences occur at smaller values of $a \sim 0.99$, close

to the separatrix. This is due to our mapping $z(a)$, as the majority of grid points are reserved for higher spins.

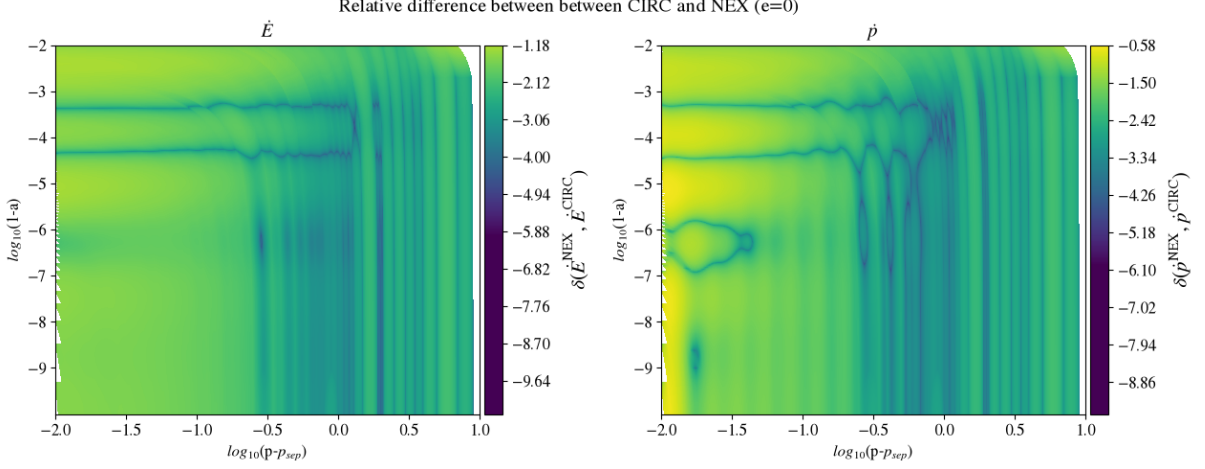


Figure 3.3: Here the relative differences between the interpolated circular data, and interpolated near-extremal data are plotted in the $p - a$ plane. Left: $\dot{E}$. Right: $\dot{p}$ ($= \dot{r}$).

We can conclude from this analysis, that while the behaviour far from the separatrix agrees with high-precision, more data points may be required closer to the separatrix to achieve better agreement with other models/datasets.

3.4 Amplitude Validation

Here, only the amplitudes from FEW’s eccentric model are compared to our interpolated amplitudes. This decision is made out of computational convenience, as FEW stores amplitudes in the form of Eq. (2.49), and so to compare the amplitudes of the circular data, the spin-weighted spheroidal harmonics must also be computed. As a result, we limit ourselves here to comparing the interpolated amplitudes of the near-extremal model with FEW’s eccentric model.

It is important to note that unlike the fluxes, whose errors accumulate over the entire course of the inspiral, the amplitude error does not. Thus, small errors in the amplitude are not as significant, whereas the same error in the flux would be far more impactful [24]. Despite this fact, due to the rapid variation of the amplitudes near the horizon, the amplitudes are evaluated on a denser grid $(N_u, N_w, N_z) = (17, 17, 17)$, as it was necessary to resolve unphysical behaviour of the waveform near the separatrix. Due to the relatively high computational cost of such a grid, the number of modes included was limited, with $(\ell_{\max}, m_{\max}, n_{\max}) = (8, 8, 92)$.

The relative difference of Eq. (3.7) is used again to compare the amplitudes. The results of the comparison are shown in figure 3.4 below.

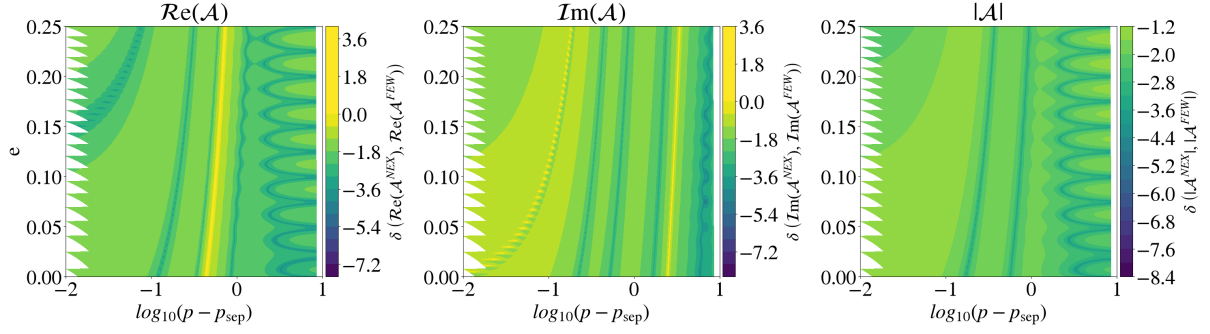


Figure 3.4: Plotted are the relative differences of the real (left) and imaginary (centre) parts of the complex amplitudes, as well as the absolute value of the entire complex amplitude (right), all in the $p - e$ plane, for $a = 0.99$.

The regions where the relative difference increases are where the amplitudes change signs, resulting in the divergence of Eq. (3.7). The real and imaginary amplitudes agree well beyond $p - p_{\text{sep}} \sim 1$, with a relative difference of $< 10^{-2}$. The absolute magnitude shows similar agreement here. All plots show an increase in agreement for higher values of eccentricity, which is likely caused by the separation of the separatrix and event horizon being greater for increased eccentricity.

Overall, the amplitudes of the two models agree with a small enough relative difference for us to move on to compare the trajectories and waveforms.

3.5 Trajectory and Waveform Comparison

We can now test how the individual differences in amplitudes and fluxes affect the waveforms. This will be done by generating a waveform with both models, using the same system parameters. Note that more sophisticated comparisons could be conducted by calculating mismatches of the waveforms, as was done in Ref. [24]. We opt instead for a simpler analysis for brevity, and leave the rigorous comparisons of the models for future work.

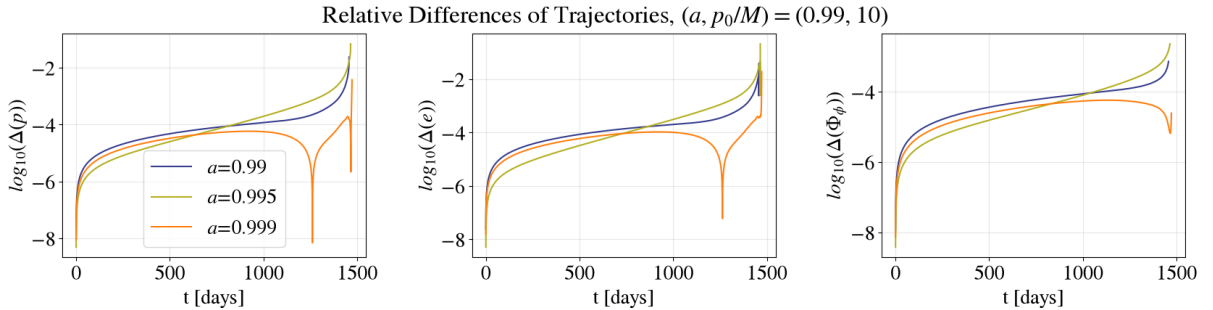


Figure 3.5: Relative difference between orbital parameters. Left: semi-latus rectum p , centre: eccentricity (e), and right: azimuthal phase Φ_ϕ . In these plots $\Delta(X)$ is shorthand for “relative difference in X ”.

To begin, we compare the slow evolution of the orbital parameters (p, e) , as well as the azimuthal phase Φ_ϕ . We compare three inspirals, for varying values of spin parameter, and fix $e_0 = 0.2$, and $p_0/M = 10$. This allows for a long inspiral of 4 years. The results are plotted in figure 3.5.

The difference is smallest in the case of $a = 0.999$, and would likely gain accuracy as the spin parameter grows. This is a direct result of the mapping from Eq. (3.6) which puts far more data points in the near-extremal region.

Similar analysis of varying e_0 yielded no significant results, with very little relative difference being exhibited.

When comparing waveforms, the phase difference that accumulates throughout the inspiral can be observed after just ~ 5 days for the system displayed in figure 3.6. The waveforms are entirely out of phase by day 15.

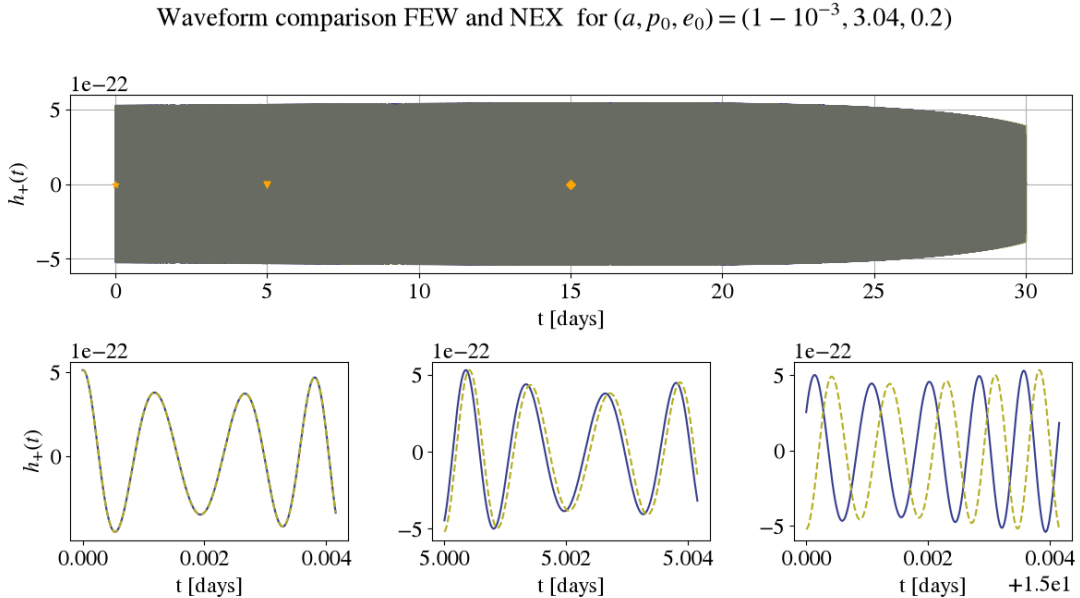


Figure 3.6: Here we compare the waveforms of FEW’s eccentric model, and our near-extremal one, at $a = 0.999$. Top: The full waveforms. FEW’s is plotted with transparency, such that any difference in the waveform shape could be seen. Bottom: Enhanced images of the waveform to examine the WF on the orbital timescale.

These comparisons of the interpolated functions, as well as the final trajectory and waveforms give us confidence that on a broad level, the WF model is working correctly. We also conclude that a denser grid would decrease the differences between the near-extremal model and FEW’s eccentric model.

Chapter 4

Results and Analysis

Every EMRI system presented here has 8 parameters:

$$\boldsymbol{\theta} = \{a, p_0, e_0, M = 10^6 M_\odot, \mu = 10 M_\odot, D = 1 \text{Gpc}, \Phi_{\phi,0} = \pi/3, \Phi_{r,0} = \pi/3\}, \quad (4.1)$$

where a, p_0, e_0 are the only parameters changed, unless stated otherwise. This choice is made simply because the main focus of these results is on studying the impact of spin and eccentricity on the trajectory of the EMRI. The variation of the other parameters is left for future work.

Each simulation is stopped at a distance from the separatrix of $p_{\text{stop}} = p_{\text{sep}} + 2\Delta p_{\text{min}}$ to avoid the edge of the parameter space, where unphysical effects induced by errors are common.

4.1 Evolution of Orbital Parameters

Upon providing the initial parameters, the ODEs in Eqs. (2.45) and (2.46) are solved numerically within FEW. To begin, we evaluate the trajectory with initial conditions ($a = 1 - 10^{-11}, p_0 = 9.4M, e_0 = 0.4$). The result is plotted in figure 4.1. The inspiral trajectory follows that of a usual EMRI until the end of the inspiral. Here the near-horizon behaviour described in section 2.5 and Refs. [29, 30] becomes apparent. The characteristic decrease in GW power is evident in the final ~ 44 hours of the inspiral, where p stops decreasing almost entirely as the end of the inspiral is reached. The azimuthal frequency of the secondary CO's orbit approaches the extremal horizon frequency of the primary of $1/2M$, a “smoking gun” characteristic of an EMRI with a near-extremal primary, with a final value of $\Omega_{\phi,\text{final}} \simeq 0.49/M$. The radial frequency rapidly decreases towards the end of the inspiral, and approaches a final value of $\Omega_{r,\text{final}} \simeq 0.00156/M$.

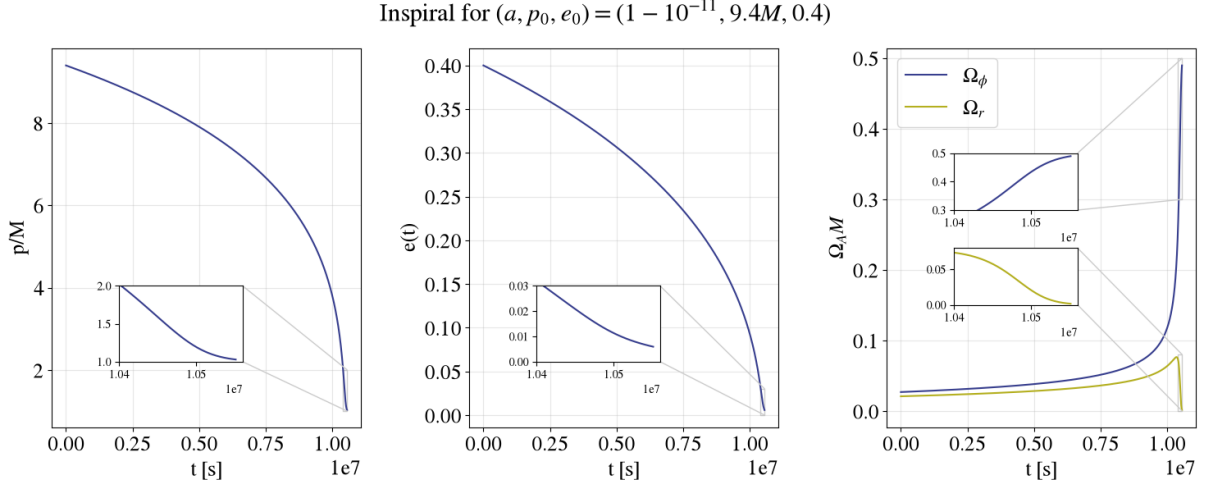


Figure 4.1: Left: the evolution of the semi-latus rectum $p(t)$ (with $p_0 = 9.4M$) as the secondary inspirals, centre: evolution of eccentricity, and right: the radial frequency Ω_r and the azimuthal frequency Ω_ϕ . Each plot has an enhanced image of the final ~ 44 hours of the inspiral.

The behaviour of the orbital parameters towards the end of the inspiral becomes clearer when plotting their rates of change, $\dot{p}$ and $\dot{e}$. Each of these is plotted in figures 4.2 and 4.3 respectively, along with the evolution of the associated parameters. We generate 7 inspirals each with $p_0 = 3.5M$ and $e_0 = 0.4$, with $a = 1 - 10^{-n}$ where $n \in \{2, 3, 4, 5, 8, 11, 12\}$ to compare how deep into the near-horizon dampening regime each inspiral goes with increasing values of the primary MBH's spin parameter.

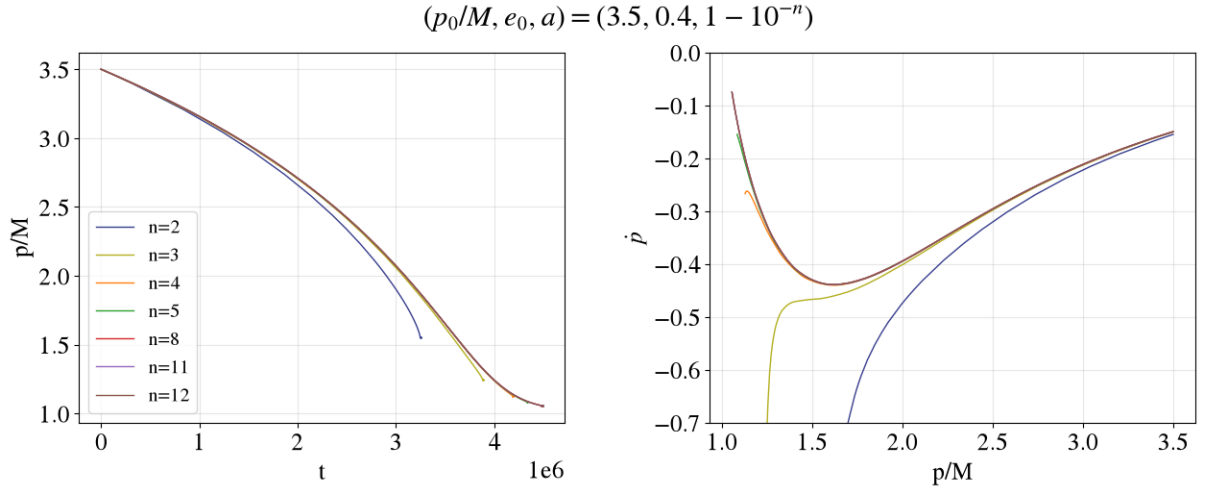


Figure 4.2: Left: The evolution of $p(t)$ and right: The evolution of $\dot{p}$ with respect to p . Each with varying values of a .

Focussing first on the behaviour of p and $\dot{p}$ in figure 4.2, this behaviour can be directly compared to what has already been found in Ref. [30], which shows a similar plot. The higher the primaries spin parameter is, the farther into the dampening regime the sec-

ondary enters, as expected as the separatrix approaches the horizon in the limit of $a \rightarrow 1$.

As shown in Ref. [56], and mentioned in figure 3.2, there exists a critical value p_{crit} for eccentric orbits at which the sign of $\dot{e}$ turns from negative to positive. A known result of this is an up-tick in eccentricity close to the separatrix as shown in Ref. [80]. This value of p_{crit} is known to move towards the separatrix as the spin parameter of the primary BH increases. Therefore, $\dot{e}$ for EMRIs with a near-extremal primary remains negative very close to the separatrix. As a result, eccentricity is radiated away for the entire inspiral, resulting in small values for final eccentricity. This effect can be observed in figure 4.3.

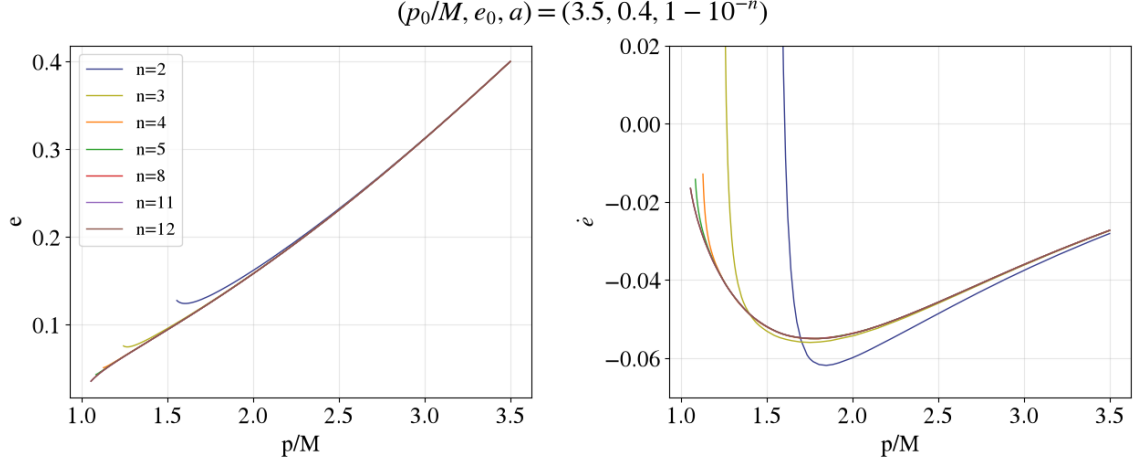


Figure 4.3: Left: The evolution of $e(t)$ and right: The evolution of $\dot{e}$ with respect to p . Again, each with varying values of a

The same quantities as above are plotted below, but for varying eccentricities and a fixed primary spin parameter of $a = 1 - 10^{-12}$

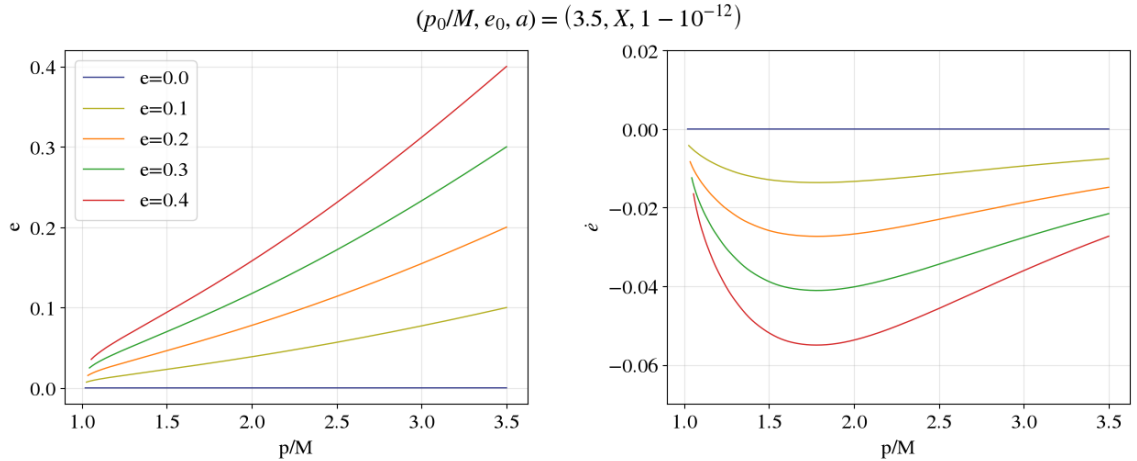


Figure 4.4: Left: The evolution of $e(t)$ and right: The evolution of $\dot{e}$ with respect to p . Each with varying values of e_0 .

It is from figure 4.4 here that the eccentricity is radiated away with great efficiency

independently of the starting value of e_0 due to the corresponding increase in $|\dot{e}|$. We now end our exploration of the trajectories produced by the inspiral generator, and turn our attention to the waveforms that they produce.

4.2 Waveforms

We provide the near-extremal WF generator with the initial parameters of $(a = 1 - 10^{-8}, p_0 = 7M, e_0 = 0.4)$ to generate a 426-day-long EMRI waveform. This took 49 seconds to generate *without* GPU acceleration. The resultant WF is plotted in figure 4.5. From the full WF, we can observe the behaviour of the WF over the full inspiral timescale $T_{\text{inspiral}} = 426$ days. The amplitude increases as would be expected of a usual WF, but at day ~ 408 , the amplitude begins to decrease and displays the “smoking gun” dampening described in Refs. [30, 29]. The snapshot waveforms inform us about how the mode content and frequency of the waveform evolve. The modulation due to eccentricity decreases as the inspiral evolves, and by the end of the inspiral, the eccentricity reached $e_{\text{final}} = 0.0098$, resulting in only very small modulations in amplitude, and no discernable modulations in WF frequency.

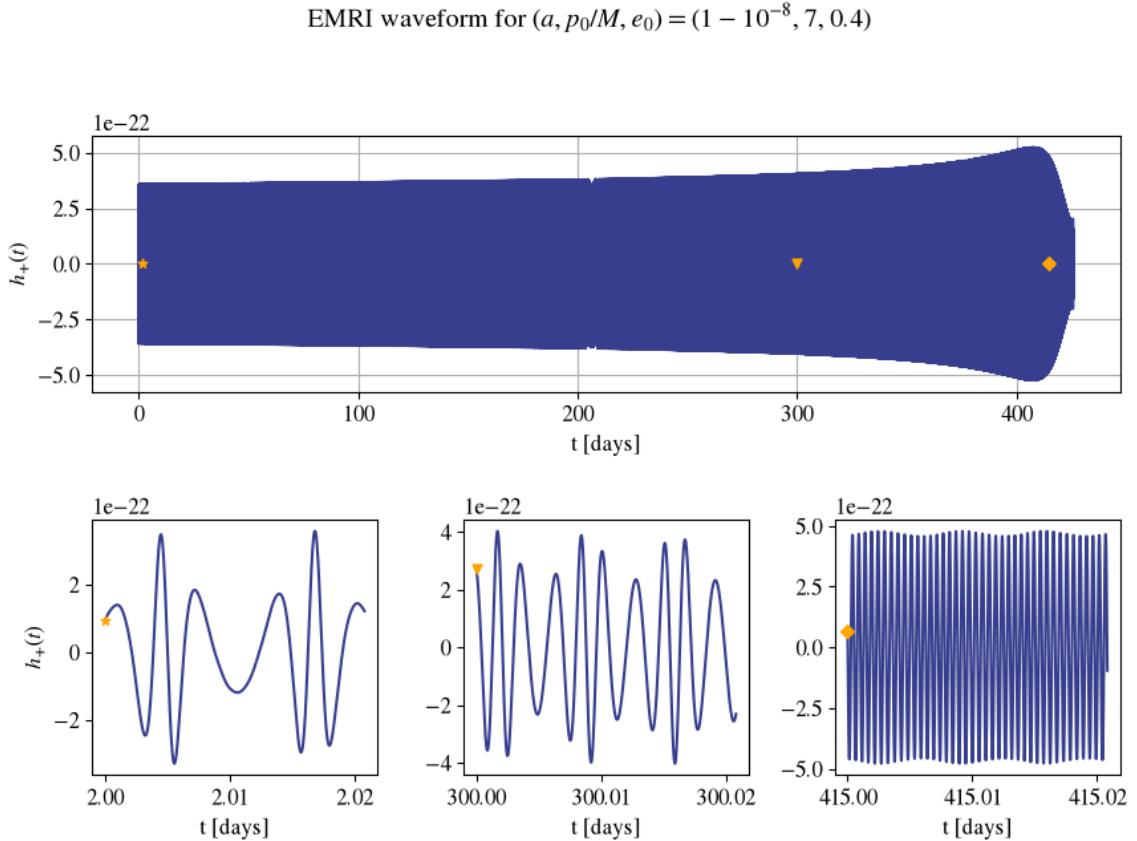


Figure 4.5: Top: full 426-day-long waveform. Bottom: enhanced images of the same waveform, at the times indicated by the symbols.

The behaviour of the frequency of the WF can be further investigated with a spec-

rogram. We generate a waveform that lasts only 60 days, with initial parameters ($a = 1 - 10^{-10}$, $p_0 = 3.75M$, $e_0 = 0.4$). The spectrogram produced by this waveform is displayed in Figure 4.6. The frequencies in the WF can be seen to be initially disjointed as a result of the initially high eccentricity. This modulation decreases rapidly as the eccentricity is efficiently radiated away, and the frequency of the radial motion subsequently plunges. All frequencies tend towards a single value as eccentricity is radiated away. The strength of each frequency is indicated by the colour bar, where the strengths of the individual frequencies at any given time are expressed with the short-time Fourier transform [81].

The frequency that is reached by the end of the inspiral is ~ 0.032 Hz. This value agrees with Eq. (2.65), as the horizon frequency for an extremal MBH with a mass of $10^6 M_\odot$. The same system can also be plotted in the frequency domain against the LISA sensitivity curve, plotted in figure 4.7. The frequencies where the signal is strongest occur at the bottom of the LISA sensitivity bucket.

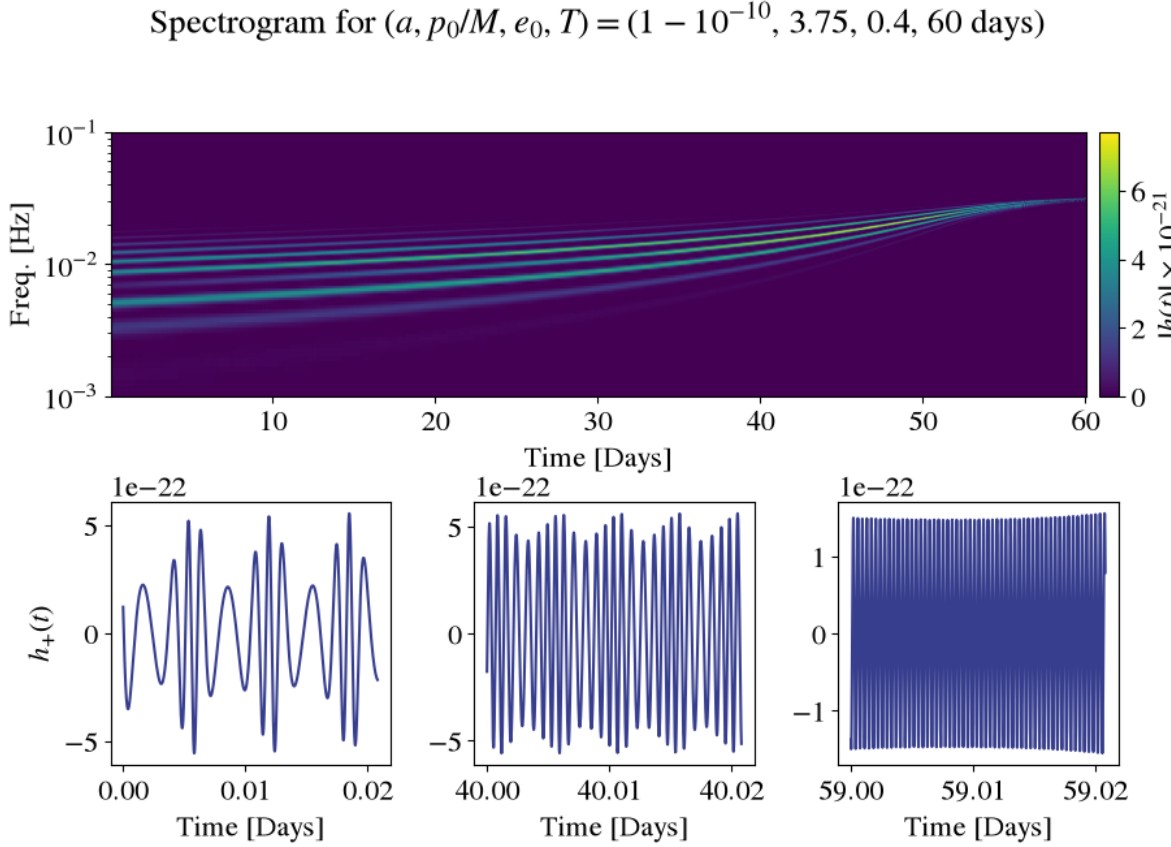


Figure 4.6: Top: spectrogram of a WF signal. Bottom: snapshots of the same WF at $T=(0, 40, 59)$ days, each lasting 30 minutes.

The LISA sensitivity curve is plotted in figure 4.7 (as is plotted in figure 2.2), along with the discrete Fourier transform (DFT) of the waveform. The DFT is computed with Numpy's FFT function, which quantifies the power of the GW signal. The frequency of

0.032 Hz is indicated by the dotted line.

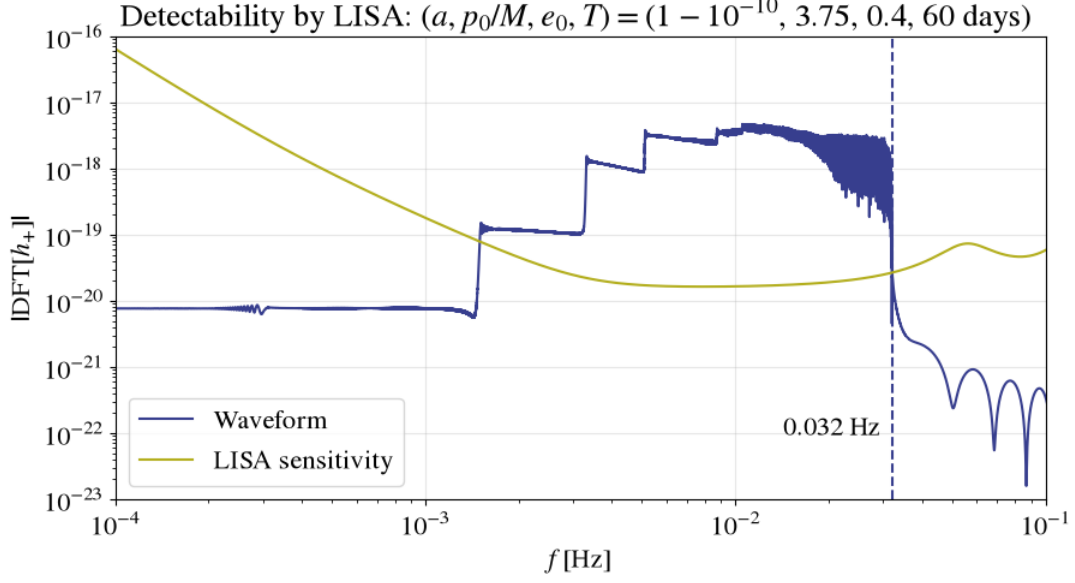


Figure 4.7: Plotted is the LISA sensitivity curve, and the DFT of the WF.

The SNR (Eq. (2.70)) of the signal plotted in figures 4.6 and 4.7 was calculated using the discretised noise-weighted inner product, Eq. (2.74), and was found to be $\simeq 54.5$.

Finally, we comment on the “inverse zoom-whirl” behaviour of eccentric EMRIs near the horizon. To examine this phenomenon, we generate a short inspiral close to the separatrix, with $p_0 = p_{\text{sep}} + 0.0245M$, and choose an eccentricity of $e = 0.2$. We plot the trajectory of the EMRI in the $x - y$ plane (viewed from above), and the associated waveform.

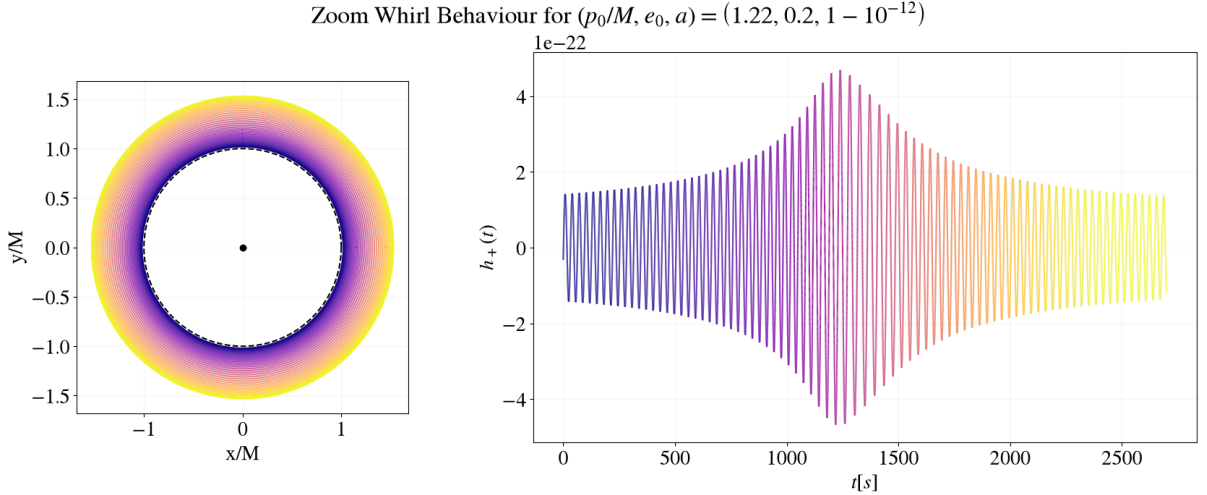


Figure 4.8: The same style of plot as figure 2.3. Left: trajectory of the EMRI in the $x - y$ plane, and right: the associated waveform. The event horizon of $r_+ \simeq 1$ is plotted as the dotted black circle.

The EMRI snapshot plotted in figure 4.8 starts at periastron at $r_p \simeq 1.0167M$, and moves out to apastron at $r_a \simeq 1.525M$. The colour of the plots tracks the distance from the horizon. At apastron, the waveform amplitude can be seen to be suppressed. This is caused by a sizeable portion of the gravitational radiation plunging down the horizon of the primary BH, and thus does not contribute to the waveform amplitude. As the CO moves out of the near-horizon regime, the amplitude increases, as the flux can now escape to infinity with greater efficiency while also being very close to the primary BH. The amplitude then decreases again as the CO moves to apastron as expected.

4.3 Constraining the Spin Parameter

We now seek to utilise the near-extremal WF model to extend the results of Ref. [30] to take into account the effects of eccentricity and initial radial phase when estimating parameter precisions with Eq. (2.73).

We compute the Fisher information matrix via Eq. (2.71), using the five point stencil finite difference scheme of Eq. (2.75) and the inner product of Eq. (2.74). With the eight parameters listed in Eq. (4.1), each Fisher matrix is a symmetric 8×8 matrix. A total of 32 waveforms must be computed per Fisher matrix, which without GPU hardware can be a computationally expensive task. As such, to garner results within the time frame of this project, each Fisher matrix was computed with an 0.4 year-long inspiral, and the value of p_0 was selected using FEW's `get_p_at_t` function. A sampling interval of $dt = 50$ s was chosen to allow for the computations to be reasonably inexpensive.

We calculate the Fisher matrices over the $a - e_0$ parameter space to test the effects of increasing eccentricity on the precision of the spin parameter, obtained with Eq. (2.73). This allows us to generalise the results of Ref. [30], by taking into account the eccentricity and initial radial phase parameters in the calculation of the precisions.

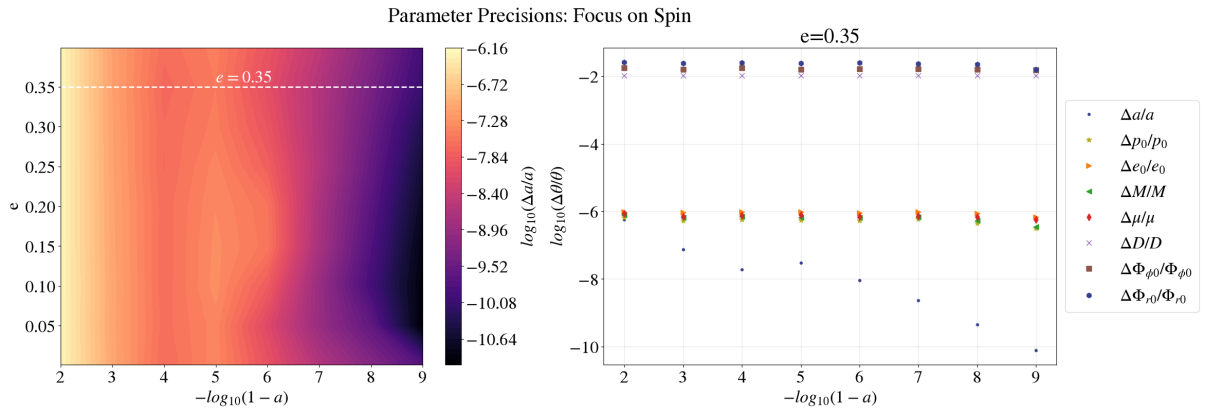


Figure 4.9: Left: the percentage precision $\Delta a/a$ is plotted in the $a - e_0$ plane, and the $e = 0.25$ line is indicated. Right: fixed $e_0 = 0.35$, and varying a the percentage precision of all 8 parameters are plotted against increasing spin.

Figure 4.9 shows a very weak dependence on eccentricity for the precision of the spin measurement. The results of Ref. [30] are not affected by the inclusion of the additional parameters e and $\Phi_{r,0}$. We also plot the percentage precision of the eccentricity in the $a - e_0$ plane, and see that the precision of the initial eccentricity is independent of the spin.

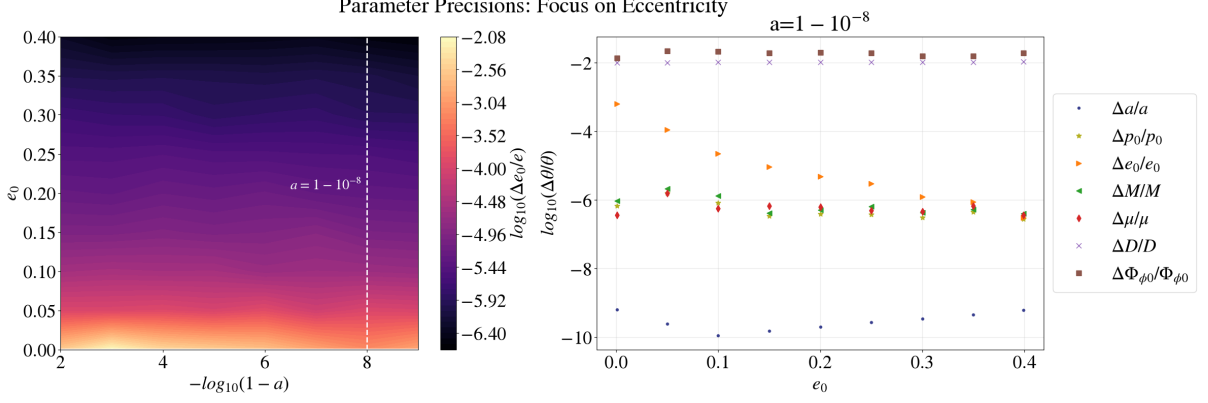


Figure 4.10: Left: the percentage precision $\Delta e_0/e_0$ is plotted in the $a - e_0$ plane, and the $a = 1 - 10^{-8}$ line is indicated. Right: Fixed $a = 1 - 10^{-8}$, and varying e_0 the percentage precision of 7 parameters are plotted against increasing spin, not including $\Phi_{r,0}$.

4.4 Discussion

When testing the behaviour of the trajectory model for EMRIs about near-extremal primaries, the expected behaviours presented in Refs. [29, 80] were observed. With a $\Delta p_{\min}$ of 10^{-2} , the trajectories do not reach as far into the near-horizon regime as they otherwise would, as illustrated in the right panel of figure 4.2. Evaluating the fluxes closer to the separatrix would allow for the full exploration of the near-horizon behaviour. This would allow for the full evolution of the azimuthal frequency, which would reach values even closer to that of $1/2M$.

Another important feature that a full exploration of the near-horizon regime would further uncover, is the behaviour of p_{crit} . As shown in figures 4.3 and 4.4, due to the moving of p_{crit} towards the separatrix in the near-extremal case, p_{crit} is not reached for inspirals about MBHs with spins higher than $a = 1 - 10^{-4}$. This allows for the eccentricity to reach very low values, for example $e_{\text{final}} \simeq 0.006$ for the inspiral plotted in figure 4.1. It would be interesting to see how much the up-tick of eccentricity would affect these final value of eccentricity, something that could only be observed with lower values of $\Delta p_{\min}$.

We comment now on how these results could aid in improving future generations of this model, or similar models. The efficiency of the circularisation in the near-extremal limit suggests that less data points are needed for eccentricities of $e \gtrsim 0.1$ when $p_0 = 3.5M$

and $a > 1 - 10^{-4}$. With this in mind, a new mapping $w(e)$ could take the form of

$$w = \frac{e}{S(a, p)}, \quad (4.2)$$

where $S(a, p)$ is a function that tapers the density of grid points in e to be more focussed on the small eccentricity range as a increases, and p approaches the separatrix. To determine the maximum value of e that would be needed, going closer to the separatrix would be necessary, to gauge the magnitude of the up-tick in eccentricity. This would massively decrease the computational cost of generating the flux and amplitude data grids as points with high eccentricity, close to the separatrix represent the highest computation time, as demonstrated in figure 3.1. A scenario in which the alterations to the w mapping suggested above would not be useful, is the investigation of The “inverse zoom-whirl” behaviour. To observe this behaviour, higher values of eccentricity must be available very close to the separatrix.

Despite the inclusion of eccentricity, the results from the waveform generation also are in line with the characteristic behaviour of quasi-circular EMRIs with near-extremal primaries. This is due to the advance in p_{crit} towards the separatrix and its impact on the final eccentricity of an inspiral. The characteristic dampening of the waveform amplitude is exhibited at the later stages of the inspiral as plotted in figure 4.5, and the frequency modulation due to eccentricity decreases as the system circularises, as can be seen in figures 4.6 and 4.8. The final frequency that the waveform reaches is predicted accurately by Eq. (2.65).

The efficiency of the waveform generation is worth noting. While we do not conduct a rigorous check of generation times across the parameter space, we will quote some figures at the edges of parameter space. A year long waveform with initial parameters of $(a = 1 - 10^{-5}, p_0 = 6.77M, e_0 = 0.05)$ took a time of 41 seconds to generate. The same system with the initial eccentricity replaced with $e_0 = 0.4$ took 68 seconds to generate. The same system again, but now with a primary MBH spin parameter of $a = 1 - 10^{-12}$ crashed the kernel before completing. The cause of this crash has not yet been investigated. The efficiency of waveform generation could be optimised across the parameter space in future work, and could also be increased by performing the computation on a GPU, or on a high-performance computing cluster.

The results of our Fisher matrix analysis shows the same qualitative behaviour as the results of Ref. [30]. We also extend these results to include the effects of orbital eccentricity. Our results display a noticeable *decrease* in precision of the spin parameter at $a \sim 1 - 10^{-4}$. Our results also show slight variation in the precision at different values of initial eccentricity, especially noticeable at the edge of the data grids at $e \simeq 0$ and $e \simeq 0.4$, where the precision of the spin parameter is seen to decrease. These effects would require further investigation, with re-computation of the Fisher matrices on a denser grid, with a smaller sampling interval dt , and for longer inspiral times. The results could also be tested with a Markov Chain Monte Carlo (MCMC) analysis, as done in Ref. [30]. The results involving similar analysis of the precision of the initial eccentricity measurement

seem more conclusive, with only very small variations in the precision with respect to increasing the spin parameter.

Any data analysis that requires the generation of a repertoire of waveforms can be readily made far more efficient by the means of GPU acceleration, which FEW's architecture easily allows for should the hardware be available.

Chapter 5

Conclusions

5.1 Main conclusions

The primary goal of this project was to leverage pre-existing state-of-the-art WF generation tools to build an open-source WF generator that covers the near-extremal region of the parameter space of the primary MBH, while also allowing for a non-zero orbital eccentricity of the secondary CO. This goal was achieved, and the waveform model that was produced during this project allows for a spin of the primary MBH as high as $a = 1 - 10^{-12}$ and an orbital eccentricity of the secondary as high as $e = 0.4$. This waveform generator can now be used by researchers to conduct further studies of EMRIs with a near-extremal primary.

The first component of the WF model is the inspiral generator. This module allows for the slow evolution of the orbital parameters p and e , and the fundamental frequencies Ω_ϕ and Ω_r to be examined over the course of the inspiral. This allowed for the exploration of the near-horizon behaviour of the fluxes $\dot{p}$ and $\dot{e}$. In doing so, we demonstrated the well-known result of Ref. [56], that the inspirals circularises to near-zero eccentricities as a result of the rate of change in eccentricity remaining negative over the course of the inspiral. A consequence of this is that the “smoking gun” of quasi-circular EMRIs discussed in Ref. [29] is applicable to inspirals that begin with initial eccentricity, provided the inspiral lasts long enough for circularisation to take place. Specifically, the azimuthal frequency of the orbiting CO approaches a value of $1/(2M)$, the horizon frequency of an extremal BH. Simultaneously, the detector-frame waveform frequency approaches $f_{\text{detector}} \simeq 3.2 \times 10^4 (M_\odot/M) \text{Hz}$, which is expected for quasi-circular inspirals, as shown in Ref. [29].

The waveform model was then utilised to extend the results of Ref. [30] to include the effects of the two additional parameters of the initial orbital eccentricity and the initial radial phase. We demonstrated that the precision of the spin parameter measurement that LISA could achieve remains considerably higher for near-extremal systems than moderately rotating ones with an improvement of ~ 4 orders of magnitude, and our results suggest that this is independent of initial eccentricity.

The entire WF model was built within the pre-existing architecture of FEW, which demonstrates the versatility and ease of access of the codebase, despite its depth and sophisticated modelling methods. In a relatively short project, we have built a full waveform generator and used it to study the near-horizon regime of near-extremal MBHs, generating original results. Without the modular framework of waveform generation codes such as FastEMRIWaveforms, as well as high-quality data that can be readily produced using tools such as PYBHPT, this project would likely not have been possible a decade ago. This showcases the remarkable progress in the field of EMRI waveform modelling. This rapid progress is likely to continue, with LISA being due to launch in 2035 [17].

5.2 Future Work

Firstly, as discussed, the accuracy of the model can be readily improved with the inclusion of more flux and amplitude data. This may also improve the efficiency of the model, as instabilities are removed. As well as a denser grid, a wider grid that includes values of p closer to the separatrix would also be valuable in investigating the near-horizon behaviour of the waveforms.

To validate the results found by our Fisher matrix analysis, a Bayesian analysis using Markov Chain Monte Carlo techniques could be conducted. Our results are generally consistent with the quasi-circular results of Ref. [30], but still require a full Bayesian analysis to verify the results that include eccentricity. With the waveform model built in this project available, these results would be readily obtained.

The inclusion of orbital inclination of the secondary BH would be informative. This would allow for the effects of the inclusion of the inclination parameter x_I , and initial polar phase parameter $\Phi_{\theta,0}$ on the precision estimates to be analysed. A circular precessing waveform model that allows for orbital inclination is under development in FEW. The extension of this model to the near-extremal regime would be a near-identical process to the tasks described in Appendix B.

A full treatment of the inspiral to plunge about a near-extremal MBH, as studied in [57] would represent a full WF model. This would come with natural computational difficulties, as the fluxes near the separatrix become increasingly expensive to compute as shown in figure 3.1.

Appendix A

Gravitational Wave Theory

Linearised Gravity

To begin our discussion, we will cover the basics of GW theory, developing how GWs manifest in flat spacetime, such as (approximately) the spacetime on/around Earth. This discussion will be primary based on the pedagogical works [31, 5], where *linearised gravity* is discussed in greater detail than it is here. In a flat spacetime described by the Minkowski metric $\eta_{\mu\nu}$, we can add a small deviation from flatness as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (\text{A.1})$$

With this perturbed metric, we will construct the Einstein tensor $G_{\mu\nu}$, which satisfies the Einstein field equations,

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu}, \quad (\text{A.2})$$

where $R_{\mu\rho\nu}^\rho = R_{\mu\nu}$ is the Ricci tensor, $R = R^\nu_\nu$ is the Ricci curvature scalar, and $T_{\mu\nu}$ is the stress energy tensor, which embodies all information about matter and energy in our spacetime. Thus In a vacuum, $T_{\mu\nu} = 0$, the Einstein tensor satisfies the vacuum field equations

$$G_{\mu\nu} = 0. \quad (\text{A.3})$$

Our goal is to use Eq. (A.1) in the above field equation, to find how the perturbation behaves. To start towards this goal, first note the following expressions for the Riemann tensor, and Christoffel components,

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\mu\sigma}(\partial_\rho g_{\sigma\nu} + \partial_\nu g_{\sigma\rho} - \partial_\sigma g_{\rho\nu}) \quad (\text{A.4})$$

we can use equation (A.1) in the vacuum field equations, keeping only up to first order terms of $h_{\mu\nu}$, to acquire

$$\partial_\rho \partial_\nu h^\rho_\mu + \partial^\rho \partial_\mu h_{\nu\rho} - \square h_{\mu\nu} - \partial_\mu \partial_\nu h - \eta_{\mu\nu} \partial_\rho \partial^\rho h^\rho_\sigma + \eta_{\mu\nu} \square h = 0, \quad (\text{A.5})$$

where $h = h^\nu_\nu$ and $\square = \partial_\nu \partial^\nu$. This is difficult to solve, but it can be simplified by the adoption of the trace-reversed metric perturbation, followed by a gauge (coordinate)

transformation. First, we introduce the trace-reversed perturbation: $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$, which can be rearranged to garner an expression for $h_{\mu\nu}$, and thus plugged back into equation (A.5) to get

$$\partial_\rho \partial_\nu \bar{h}^\rho{}_\mu + \partial^\rho \partial_\mu \bar{h}_{\nu\rho} - \square \bar{h}_{\mu\nu} - \eta_{\mu\nu} \partial_\rho \partial^\rho \bar{h}^\rho{}_\sigma = 0. \quad (\text{A.6})$$

This expression can be simplified even further by a suitable choice of gauge

The Lorentz Gauge

Before specialising to a specific gauge, consider an infinitesimal gauge transformation

$$x^\mu \rightarrow x'^\mu = x^\mu + \varepsilon^\mu(x^\nu), \quad (\text{A.7})$$

where $|\varepsilon| \ll 1$ is any infinitesimal vector field. This is just a coordinate transformation, and thus the metric tensor — a type $(0, 2)$ tensor — transforms as

$$g'_{\mu\nu} = \frac{dx^\rho}{dx'^\mu} \frac{dx^\sigma}{dx'^\nu} g_{\rho\sigma}. \quad (\text{A.8})$$

$$= (\delta^\rho_\mu - \partial_\mu \varepsilon^\rho) (\delta^\sigma_\nu - \partial_\nu \varepsilon^\sigma) (\eta_{\rho\sigma} + h_{\rho\sigma}) \quad (\text{A.9})$$

$$\simeq \eta_{\mu\nu} + h_{\mu\nu} - 2\partial_{(\nu} \varepsilon_{\mu)} \quad (\text{A.10})$$

$$= \eta_{\mu\nu} + h'_{\mu\nu}, \quad (\text{A.11})$$

where in the first line we use the inverse gauge transformation $x^\mu = x'^\mu - \varepsilon^\mu(x^\nu)$, and on the third line we only keep first order terms of ε_μ and $h_{\mu\nu}$ (treating cross terms of each as 2nd order in either), and in the final line we introduced the gauge-transformed metric perturbation $h'_{\mu\nu} = h_{\mu\nu} - 2\partial_{(\mu} \varepsilon_{\nu)}$. We also make use of the notation,-

$$A_{(\mu\nu)} = \frac{1}{2}(A_{\mu\nu} + A_{\nu\mu}). \quad (\text{A.12})$$

The trace-reversed perturbation under the gauge transformation takes the form

$$\bar{h}'_{\mu\nu} = h'_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h', \quad (\text{A.13})$$

$$= h_{\mu\nu} - 2\partial_{(\mu} \varepsilon_{\nu)} - \frac{1}{2}\eta_{\mu\nu}\eta^{\rho\sigma}(h_{\rho\sigma} - 2\partial_{(\rho} \varepsilon_{\sigma)}), \quad (\text{A.14})$$

$$= \bar{h}_{\mu\nu} - 2\partial_{(\mu} \varepsilon_{\nu)} + \eta_{\mu\nu}\eta^{\rho\sigma}\partial_{(\rho} \varepsilon_{\sigma)}, \quad (\text{A.15})$$

$$= \bar{h}_{\mu\nu} - 2\partial_{(\mu} \varepsilon_{\nu)} + \eta_{\mu\nu}\partial^\rho \varepsilon_\rho, \quad (\text{A.16})$$

where in the second line, we use the definition of the gauge-transformed metric perturbation, as well as its trace, and in the third line we used the definition of the trace-reversed metric perturbation. To simplify equation (A.6), it would be useful if we could set $\partial^\sigma \bar{h}_{\sigma\mu} = 0$, and thus this is the condition we would like to fulfil with out gauge transformation, i.e. choose a vector ε_μ , such that this condition is true. To find such a vector, let's look at

$$\partial^\mu \bar{h}'_{\mu\nu} = \partial^\mu (\bar{h}_{\mu\nu} - 2\partial_{(\mu} \varepsilon_{\nu)} + \eta_{\mu\nu}\partial^\rho \varepsilon_\rho). \quad (\text{A.17})$$

After performing some tensor algebra, this can be shown to be

$$\partial^\mu \bar{h}'_{\mu\nu} = \partial^\mu \bar{h}_{\mu\nu} - \square \varepsilon_\nu. \quad (\text{A.18})$$

The Lorentz Gauge condition is thus fulfilled by choosing $\varepsilon(x^\nu)$ such that

$$\square \varepsilon_\mu = \partial^\nu \bar{h}_{\mu\nu} \quad (\text{A.19})$$

This equation is always solvable, as it is a wave equation, and thus we are always free to choose a coordinate transformation that satisfies it, thus ensuring that the Lorentz gauge is always a valid choice of gauge. We will thus enforce the Lorentz condition

$$\partial^\mu \bar{h}_{\mu\nu} = 0, \quad (\text{A.20})$$

which greatly simplifies equation (A.6) to

$$\square \bar{h}_{\mu\nu} = 0, \quad (\text{A.21})$$

which is a wave equation, the solutions to which are gravitational waves

$$\bar{h}_{\mu\nu}(\vec{x}, t) = \text{Re} \left(\int d^3k A_{\mu\nu}(\vec{k}) e^{i(\vec{x} \cdot \vec{k} - \omega t)} \right), \quad (\text{A.22})$$

where $\omega = |\vec{k}|$, and the Lorentz gauge condition requires that $k^\mu A_{\mu\nu} = 0$ where $k^a = (w, \vec{k})$.

The Transverse Traceless Gauge

In a globally vacuum spacetime that is asymptotically flat, meaning that $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$ as $r \rightarrow \infty$ (meaning that $h_{\mu\nu} \rightarrow 0$), there is further coordinate freedom to further restrict the solutions to equation (A.21). Along with the Lorentz condition, we further restrict the gauge to be purely spatial and traceless

$$h_{tt} = h_{ti} = 0, \quad (\text{A.23})$$

$$h = h_i^i = 0, \quad (\text{A.24})$$

where now Latin indices i represent the spatial parts of the metric. These constraints, alongside the Lorentz condition (A.20), is known as the ‘transverse traceless gauge’, or the TT gauge. In this gauge, GWs have only two components, h_+ and $h_\times$, which are the two waveform components that shall be referenced throughout this project.

Appendix B

Software Development

To utilise the FastEMRIWaveforms codebase to build a waveform model, I created a fork of the FastEMRIWaveforms GitHub repository. To make the downloading of FEW more convenient, I made use of Windows Subsystem for Linux (WSL). This allowed me to edit the codebase, and keep my progress synchronised online throughout the project. The FastEMRIWaveforms GitHub repository can be accessed [here](#), and my fork can be found [here](#). This branch also has an additional folder in the *examples* folder, labelled “extra”. This is where the code that was used to generate the results in this project can be found.

First, a list of what files need to be changed to implement the waveform generator. Note that the exact changes made to the codebase are available in the commit history of the `Dedicated_near_extremal_WF` branch.

1. *kerreccq.py*: The mappings between (u, w, z) and (a, p, e) are added here, as well as the grid boundaries.
2. *baseclasses.py*: A new base-class is created, `KerrEccentricEquatorial_nex`. The relevant value for the grid boundaries must be implemented.
3. *flux.py*: An interpolation class `KerrEccEqFlux_nex` is created, to allow for the interpolation of the new flux data on the new grid.
4. *ode/_init_.py*: Add the `KerrEccentricEquatorial_nex` base-class to the stock trajectory options.
5. *registry.yml*: Use Sha256 algorithm to generate a key for the data files.

These changes are all that is needed to create a trajectory model. To get a full waveform model the following additional steps are required,

1. *ampinterp2d.py*: Add a new amplitude class `AmpInterpKerrEccEq_nex` to allow for the generation of amplitudes on the new data grid, with the new data.
2. *waveform/_init_.py*: Add `AmpInterpKerrEccEq_nex`.

3. *waveform.py*: Add a `FastKerrEccentricEquatorialFlux_nex` class to allow for the summation of amplitudes generated by `AmpInterpKerrEccEq_nex`, and sum them to generate a waveform.

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